

1 Physical Primitives of Mind: Modeling
2 Consciousness via Symmetries and Landauer
3 Erasure

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6 **Abstract**

7 We present a mathematical synthesis and formal Lean 4 verifi-
8 cation demonstrating how cognitive architectures can emerge as at-
9 tractor states from established physical primitives, without introduc-
10 ing new fundamental physics. Starting from the Poincaré group of
11 spacetime symmetries and finite phase space, we construct a frame-
12 work for predictive processing. Spontaneous symmetry breaking gen-
13 erates necessary topological boundaries, and finite phase space sub-
14 jects these boundaries to Landauer’s thermodynamic erasure. Agency
15 arises when systems evolve along paths of least action, minimizing en-
16 tropy production via internal structural resonance with external per-

17 turbations. Scaling this dynamic to macroscopic levels aligns with
 18 emerging paradigms in neuroscience—specifically classical electrody-
 19 namic fields (ephaptic coupling)—and Kuramoto phase synchroniza-
 20 tion, culminating in a topological fixed point: a unified “Self.” We
 21 state the claim narrowly, because the subject matter invites a wider
 22 reading than the work supports: no new physics is derived here. Every
 23 physical result the chain invokes is established—Picard–Lindelöf, Kib-
 24 ble’s classification of topological defects, Landauer’s erasure bound,
 25 Gibbs’ inequality, Still’s bound on dissipation by nonpredictive in-
 26 formation, Barbalat’s lemma, Sakaguchi’s threshold, Banach’s fixed
 27 point. What is new is that they are stated in one formal language,
 28 their consistency mechanically checked, and the points where the chain
 29 relies on stipulation rather than derivation located exactly. The de-
 30 velopment declares no axioms and every structure carrying physical
 31 content is inhabited by a witness; the synchronization step is proved
 32 from a dynamics for initial data inside an explicit basin and assumed
 33 outside it, and the identification of a global section of the probability
 34 sheaf with the unity of experience is a stipulation, marked as such
 35 throughout rather than absorbed into the derivation. The step from
 36 a coarse-grained coupling kernel to the brain’s endogenous electro-
 37 magnetic field is likewise an empirical commitment and not a deduc-
 38 tion; we separate it from the coarse-graining theorem that precedes it,
 39 weaken it to a modulatory-but-integrative role, and give four condi-
 40 tions that would falsify it. The framework is nonetheless falsifiable in

one place that is its own and no rival's: recovery from sudden awakening should follow a delayed sigmoid rather than an exponential, because the variable gating it is astrocytic volume on a timescale of seconds to minutes rather than cortical phase realignment on a timescale of milliseconds. As a corollary, we show why rigid von Neumann architectures (e.g., GPUs) are thermodynamically disadvantaged or structurally constrained from reaching these specific continuous thermodynamic attractors.

1 Introduction

The pursuit of a physical theory of consciousness typically involves searching for the biological correlates of mind or treating subjective experience as an abstract computational algorithm. This paper asks a more fundamental question: If we assume only basic physical primitives—continuous symmetries (the Poincaré group), geometry, and thermodynamics—is the emergence of a unified, perceiving entity a natural physical attractor?

We propose a framework where the structural properties of consciousness are derived as physical attractor states. Establishing a deductive chain from basic principles, we demonstrate how continuous symmetries break to form boundaries, how those boundaries dissipate free energy as non-equilibrium steady states, and how their evolution along paths of least action results in physical resonance with their environment. This eventually scales into the macroscopic electromagnetic unity proposed by emerging field theories

63 of mind. To avoid critiques of “math theatre,” we delineate between literal
 64 physical mechanisms (e.g., thermodynamic erasure via Landauer’s principle,
 65 boundaries from spontaneous symmetry breaking) and theoretical mappings,
 66 utilizing Lean 4 formalization as our mathematical anchor. We adopt a per-
 67 spective aligned with Russellian Monism: objective measurements of macro-
 68 scopic electromagnetic fields constitute the third-person “outside,” while the
 69 system’s internal physical deformation to reach structural resonance consti-
 70 tutes the first-person intrinsic experience.

71 **What is new here, and what is not**

72 A framework paper about consciousness invites a particular suspicion, and it
 73 is better answered at the start than at the end. **No new physics is derived**
 74 **here.** Every physical result the chain invokes is established, and most of
 75 it is old: Picard–Lindelöf for the well-posedness of the coupled dynamics
 76 (Coddington & Levinson, 1955); Kibble’s classification of topological defects
 77 by the homotopy of the vacuum manifold (Kibble, 1976); Landauer’s bound
 78 on the heat cost of erasure (Landauer, 1961); Gibbs’ inequality for the relative
 79 entropy (Cover & Thomas, 2006); Barbalat’s lemma for the asymptotics of a
 80 dissipating trajectory (Barbalat, 1959); the threshold $K_c = 2D$ for the noisy
 81 mean-field Kuramoto model (Sakaguchi, 1988); the bound of Still, Sivak,
 82 Bell and Crooks relating dissipation to nonpredictive information (Still et
 83 al., 2012); the identity of Kawai, Parrondo and Van den Broeck equating
 84 dissipated work with the relative entropy between forward and time-reversed

85 ensembles (Kawai et al., 2007); and Banach’s fixed-point theorem (Banach,
86 1922). A physicist reading this paper will recognize all of it and would
87 establish the point in three paragraphs if we did not. Conceding it costs
88 nothing.

89 What is new is of a different kind, and it is a smaller claim than a theory
90 of consciousness usually makes. These results have been stated in one formal
91 language, their mutual consistency has been mechanically checked, and the
92 places where the chain relies on stipulation rather than derivation have been
93 located exactly. The value of a machine-checked chain is not that any link is
94 novel; it is that no link can quietly borrow from another, that a hypothesis
95 cannot be smuggled in as a definition, and that the seams — where physics
96 stops and interpretation begins — are fixed in the source rather than nego-
97 tiable in the prose. Section 2.1 reports what the check cost: three of five
98 original axioms each proved **False**, from one shared root cause, and none of
99 the three had been caught by reading.

100 The framework is not thereby unfalsifiable, and one of its predictions is
101 its own rather than a rival’s. Section 11 argues that recovery from sudden
102 awakening should follow a *delayed sigmoid* rather than an exponential, be-
103 cause the variable gating recovery is astrocytic volume and cerebrospinal-fluid
104 clearance, on a timescale of seconds to minutes, while the cortical rhythms
105 whose realignment is usually taken to constitute the recovery move in mil-
106 liseconds. That is a claim about functional form; it is fittable against data of
107 a kind already collected; and a purely electrical account of the same recovery

108 predicts a different form.

109 **2 Formalization in Lean 4**

110 A core contribution of this work is the verification of our deductive chain
111 using the Lean 4 interactive theorem prover. This approach builds on recent
112 efforts to formalize consciousness and brain semantics, such as the 'OnlyOne'
113 system (Scherf, 2026) and sheaf-theoretic models of neural manifolds (Inoué,
114 2026). It is necessary to explicitly distinguish which claims have been math-
115 ematically derived as theorems within the proof assistant, and which remain
116 irreducible physical postulates. Figure 1 shows the chain with the status of
117 every link marked, and Table 1 states in six rows what each link assumes.
118 The complete claim-by-claim map to Lean identifiers, which runs to several
119 pages, is Table S1 in the supplement; the main text does not depend on
120 reading it.

121 **2.1 Soundness of the formalization**

122 Formalization is only as trustworthy as the axioms it rests on, and an axiom
123 that is inconsistent silently makes every theorem downstream of it vacuous
124 while leaving the build green. We report a case of exactly this, found by
125 auditing our own development, because the failure mode is easy to reproduce
126 and we have not seen it discussed in the applied-formalization literature.

127 Earlier drafts of this work declared five axioms. Three of them were

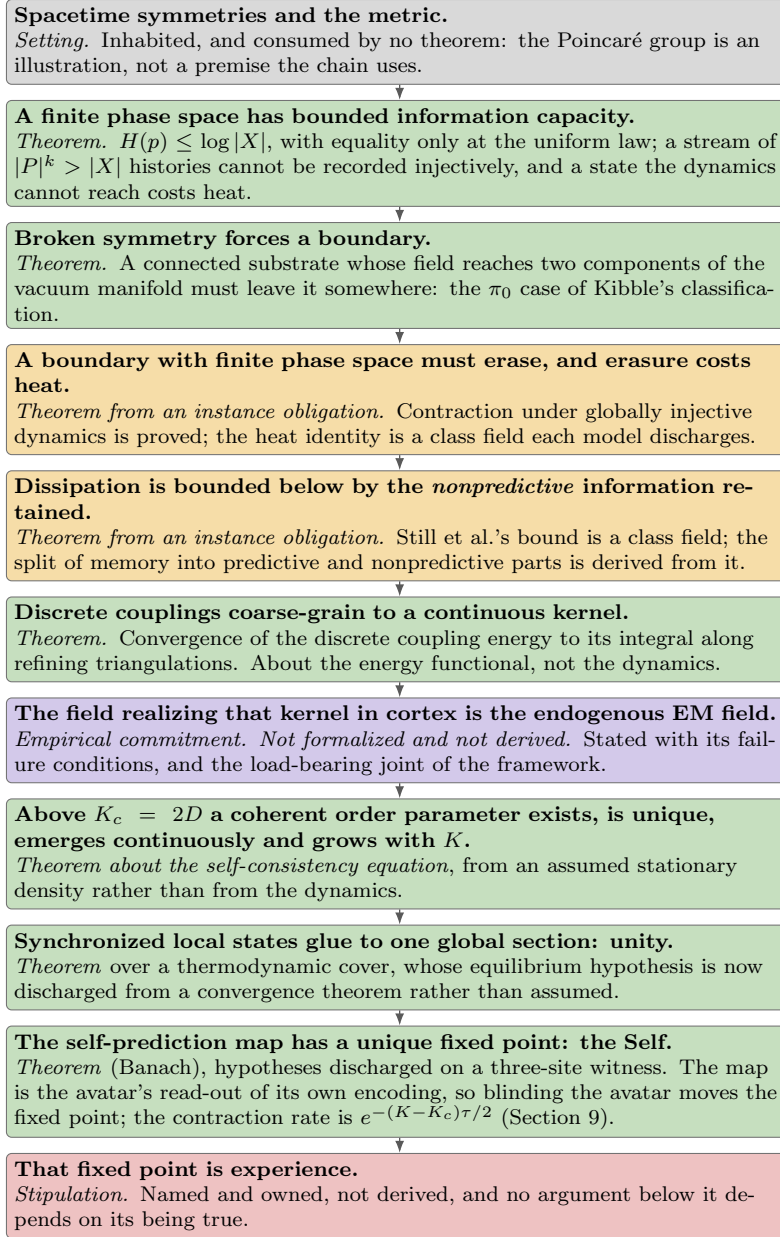


Figure 1: The deductive chain, with the status of each link marked. ■ theorem; ■ theorem resting on a class field that each model must discharge; ■ empirical commitment, neither formalized nor derived; ■ stipulation; ■ vocabulary, consumed by no theorem. The two weak links are marked rather than argued around: the identification of the coarse-grained kernel with the cortical electromagnetic field is empirical, and the final step to experience is stipulated. Table 1 states what each link assumes; the full claim-by-claim map to Lean identifiers is Table S1 in the supplement.

Claim	Status	What is assumed
A finite phase space has bounded information capacity	Theorem	Finiteness of the state space, and nothing else — no symmetry group, no metric. $H \leq \log X $, with equality only at the uniform law. On an infinite state space the step from “unreachable state” to “erasure” fails, and the failure is exhibited rather than assumed away.
Broken symmetry forces a boundary	Theorem	Substrate connected; vacuum manifold disconnected. No hypothesis mentions energy, a potential, or a symmetry.
Erasures costs heat	Theorem + class field	Global system–bath dynamics injective (Liouville). The heat identity <code>heat_eq</code> is discharged per instance, not asserted for all models.
Dissipation is bounded below by nonpredictive information	Theorem + class field	Still et al.’s bound is discharged per instance; the environment evolves by a Markov kernel the system does not act on.
Discrete coupling coarse-grains to a continuous field; that field is the cortical EM field	Theorem; then empirical	Convergence holds along regular refining triangulations, for the energy functional and not the dynamics. The physical identification is a commitment with stated failure conditions (Section 7).
Above $K_c = 2D$ a unique coherent state exists, glues into a global section, and carries a unique reflexive fixed point	Theorem	Von Mises stationary density assumed rather than derived from the Fokker–Planck operator; couplings positive and symmetric, so unfrustrated. The self-prediction map is the avatar’s read-out of its own encoding, and its Lipschitz rate $e^{-(K-K_c)\tau/2}$ is an instance obligation; the contraction — hence the fixed point — follows from $K > K_c$ and fails at or below it. The step from fixed point to experience is stipulated.

Table 1: The chain in six claims. “Class field” means a physical postulate that each model discharges for itself rather than a global axiom; the development declares **no axioms**, and `#print axioms` on any result reports only `propext`, `Classical.choice` and `Quot.sound`. Earlier drafts carried five axioms, three of them inconsistent (Section 2.1). The complete map from claims to Lean identifiers, with the scope limitation of each, is Table S1 in

128 individually inconsistent: each, on its own, proves `False`. All three shared
 129 one root cause — an axiom that constrains a symbol it does not itself bind.
 130 Concretely:

- 131 • `spontaneous_symmetry_breaking_pointwise_min` took the potential-
 132 energy functional as an *implicit* argument occurring only in its hypoth-
 133 esis `PotentialEnergy phi = V v0`. Instantiating that functional with
 134 the constant map $\lambda_. V(v_0)$ discharges the hypothesis by `rfl` and yields
 135 the conclusion for an arbitrary field. With $V(v) = v^2$, $v_0 = 0$ and $\phi \equiv 1$
 136 this proves $1 = 0$.
- 137 • `landauer_heat_eq` equated the free class field `heat_dissipation` with
 138 a bath-entropy difference *for every instance* of the class. An instance
 139 with zero heat dissipation and a bath that gains one bit gives $0 = \log 2$.
- 140 • `kl_bound_axiom` asserted $\sigma \geq D_{KL}(P||Q)/\Delta t$ for *all* P and Q , while σ
 141 is a single real fixed by a class field. Since D_{KL} is unbounded above, a
 142 point mass against a uniform distribution refutes it.

143 The remaining two axioms, `phase_invariant_periodic` and `sync_to_section_eq`,
 144 had the same shape — global assertions pinning free fields of `LocalSectionSynchronization`
 145 — and are refutable as soon as the probability presheaf admits two distinct
 146 global sections. (Both were demoted to class fields at the time; both have
 147 since been removed from the class altogether, for a separate reason given in
 148 Derivation 5.)

149 The correction is uniform, and we state it as a rule we now follow: *a*
 150 *physical postulate that mentions a class field must be a field of that class,*
 151 *never a standalone axiom quantified over all instances.* A postulate should
 152 be an obligation each model discharges, not a global claim about every model.
 153 Recast this way, the development declares no axioms at all. Two of the five
 154 became genuine theorems — pointwise vacuum minimization is now proved
 155 from $\int f = 0 \wedge f \geq 0 \Rightarrow f = 0$ a.e., and Gibbs’ inequality was already one
 156 — and the other three became instance obligations.

157 This makes non-vacuity the live question, since an uninhabitable class
 158 makes its theorems vacuous just as surely as an inconsistent axiom did.
 159 **Examples.lean** therefore constructs witnesses: a one-bit erasure system with
 160 a genuine bath that discharges **heat_eq** and reproduces $\Delta Q = k_B T \ln 2$;
 161 a perfectly resonant system discharging **kl_bound**; and a scalar field on
 162 a one-point spacetime discharging **ActionPrinciples**. Derivation 5’s glu-
 163 ing structures are witnessed too, which was until recently the largest open
 164 gap: **Examples.lean** builds a **ThermodynamicCover** over a three-site sub-
 165 strate carrying the discrete topology and its Borel σ -algebra, covered by two
 166 *overlapping* open patches whose intersection is a single shared site. The
 167 local sections are germ families of a phase-indexed finite measure $\mu_\theta =$
 168 $(1 + \cos \theta)^+ \delta_{\text{mid}}$ pushed into the sheafification of the probability presheaf; the
 169 family is provably non-constant in θ (**phaseMeasure_not_const**), so the pe-
 170 riodicity obligation of the constructor **LocalSectionSynchronization.of-**
 171 **InvariantMeasure** is discharged by an argument about $\cos$ rather than satis-

172 fied trivially, and `thermodynamic_equilibrium` is *derived* from `phase_locked_minimizes_potential`
 173 rather than assumed. `global_section_from_thermodynamics` then fires on
 174 this instance and glues the two patch-local sections into a unique global one.
 175 Two further covers answer the follow-up question of how non-degenerate such
 176 a witness can be made. `Examples.lean` §13 has a negative half worth record-
 177 ing: the phase field can be made non-constant as a function, but not as a
 178 physical state, because `thermodynamic_equilibrium` forces every instance
 179 to a phase-locked configuration. `Examples.lean` §14 is built the other way
 180 round, from two patches with independently chosen mass profiles that meet
 181 only on the site they share, and the section the gluing returns is neither of
 182 them. That is the same kind of result as `contracting_implies_const` in
 183 Derivation 6 — a proof that the witness is as strong as the class permits,
 184 rather than a witness that could be improved by trying harder.

185 What the witnesses establish is that the classes are inhabitable, so the
 186 theorems about them are not vacuous. What they do not establish is that
 187 a cortex is such an instance. In the cover above the phase field is already
 188 constant — every patch locked at $\theta = 0$ — which is precisely what makes
 189 the equilibrium obligation dischargeable; getting a physical system into that
 190 configuration is the content of the informal dynamical argument, not of the
 191 Lean development.

192 The same audit turned up a defect of a different kind, worth record-
 193 ing because it is not an axiom problem and would not have been caught
 194 by `#print axioms.mesh_refinement_convergence` — the claim that dis-

195 create edge weights converge to the continuous energy as the mesh is refined
 196 — sat in a “Conjectures” section as an unproved `def ...: Prop`. It was
 197 not merely unproved but false. Nothing required the edge regions to cover
 198 the manifold, and $\text{diam } \emptyset = 0$, so the triangulation whose edge regions are
 199 all empty satisfied the mesh hypothesis while contributing nothing to the
 200 sum, forcing $|0 - \int f| < \varepsilon$ for every $\varepsilon > 0$. Its binder was dead as well: it
 201 quantified over a triangulation and then resolved the vertex type by instance
 202 search rather than through the bound variable, so the variable was never
 203 used. `Phase2_MeshConvergence.lean` replaces it with a theorem. Conver-
 204 gence is now stated along a *sequence* of triangulations of a named region
 205 S , each required to be regular — edge regions measurable, without self-
 206 loops, disjoint across distinct edges, covering S , and anchored to the sim-
 207 plicial data in the sense that a vertex lies in the closure of every region
 208 of an edge at it. Closure rather than membership is forced: cells are dis-
 209 joint, so a shared vertex belongs to at most one of them. The manuscript’s
 210 discrete energy $\frac{1}{2} \sum_u \sum_v w(u, v) f(u)$, which samples only the first endpoint
 211 and is therefore not symmetric, is shown to equal exactly the midpoint rule
 212 $\sum_{u < v} w(u, v) \frac{1}{2} (f(u) + f(v))$ over unordered edges; without that identifica-
 213 tion, reading the double sum as a Riemann sum would be a category error.
 214 At fixed mesh the error is at most $\varepsilon \mu(S)$ with ε a modulus of continuity of
 215 the integrand, and convergence follows for uniformly continuous integrands
 216 as the mesh shrinks. The regularity conditions pull against each other, so
 217 `Examples.lean` §6 discharges them with the uniform partition of $[0, 1)$ into

218 half-open cells and checks that the approximation genuinely moves — $\frac{1}{2}$ at
 219 $N = 1$ and $\frac{1}{4}$ at $N = 2$ for the tent function $x \mapsto |x - \frac{1}{2}|$ — rather than
 220 converging trivially. What is proved is convergence, not the $O(1/N^2)$ rate
 221 the accompanying simulation measures; that would need a midpoint error
 222 term with a second derivative, which we have not developed.

223 A third defect was one of bookkeeping across files, and it is the kind that
 224 survives any number of green builds because each half is individually cor-
 225 rect. The development carried *two* Kuramoto potentials. Lyapunov descent
 226 along the dynamics, $\dot{V} = -\sum_i \dot{\theta}_i^2$, was proved for the full potential $V(\theta) =$
 227 $-\frac{1}{2} \sum_{i,j} A_{ij} \cos(\theta_i - \theta_j) - \sum_i \omega_i \theta_i$, while the characterization of the minimum
 228 — phase-locked in both directions — and everything downstream of it, in-
 229 cluding Derivation 5’s gluing, concerned the coupling term alone. Nothing
 230 chained them, and the gap was not cosmetic: `kuramoto_potential_unbounded_below`
 231 now proves that as soon as one $\omega_i \neq 0$ the full potential is unbounded below
 232 — walk θ out along ω and the bounded cosine sum cannot compensate — so
 233 it has no minimum for phase-locking to attain, and the sentence “the phase-
 234 locked state minimizes the Lyapunov potential” was, read literally, false of
 235 the functional the descent theorem was about. `Phase4_RotatingFrame.lean`
 236 closes this by formalizing the standard reduction $\theta_i \mapsto \theta_i - \Omega t$. For identical
 237 natural frequencies $\omega \equiv \Omega$, a trajectory maps to a trajectory of the zero-
 238 frequency system — which is exactly the system Derivation 5 already worked
 239 with — the two potentials coincide there, and the descent result transports
 240 onto the coupling potential whose minimizers are the phase-locked states.

241 Phase differences, phase-locking and the magnitude of the order parameter
 242 are all proved invariant under the change of frame, so nothing observable
 243 is lost in the reduction. `Examples.lean` §7 runs the chain end to end on
 244 two oscillators with a common frequency, discharging every hypothesis. The
 245 restriction is real and we state it plainly: the reduction is exact only for
 246 identical frequencies. A genuine spread of natural frequencies leaves residual
 247 detunings in the reduced system, phase-locking then requires coupling above
 248 a critical K_c , and that K_c — Kuramoto’s threshold for a spread of frequen-
 249 cies, not the noise threshold $2D$ of Section 10 — is nowhere in our Lean
 250 development. It is studied numerically, in Figure 5, as Table S1 records.

251 A fourth defect was dimensional, and it concerned the one row of Table S1
 252 that is still numerical rather than proved. `Phase8_ContinuousField.lean`
 253 defines the critical coupling $K_c = 2D$ and a predicate for being above it.
 254 That predicate compared $\int_x \int_y K(x, y)$ against $2D$ — but a double integral
 255 against an unnormalized measure carries the substrate’s mass twice over,
 256 so a constant kernel K on a substrate of total mass m contributed Km^2 .
 257 The comparison therefore depended on how large the substrate was rather
 258 than on how strongly it was coupled, and any kernel whatever could be
 259 pushed above threshold by inflating m . `Examples.lean` §9 exhibits the fail-
 260 ure concretely: on a two-site substrate carrying counting measure, a sys-
 261 tem coupled at $K = 1/2$ against a threshold of $K_c = 1$ — a factor of two
 262 *below* threshold — had an unnormalized double integral of $2 > 1$ and so
 263 satisfied the old predicate. The fix is to require the substrate’s measure to

264 be a probability measure, which is the normalization in which the mean-
 265 field derivation of $K_c = 2D$ is carried out; `mean_field_coupling_const`
 266 then gives that a constant kernel has mean-field strength exactly K , and
 267 `exhibits_phase_transition_const_iff` reduces the predicate to $K > 2D$.
 268 We flag what this does and does not do. It makes the predicate say something
 269 whose truth depends only on the physics, and §9 witnesses it on both sides of
 270 the threshold. It does not by itself prove $K_c = 2D$: the content of that value
 271 is the bifurcation of the self-consistency equation $r = I_1(Kr/D)/I_0(Kr/D)$
 272 at $K = 2D$, which is a separate question from whether the predicate is di-
 273 mensionally sound. That bifurcation is proved below, in both directions,
 274 from an assumed stationary density; deriving the density itself needs the
 275 Fokker–Planck operator and is not done.

276 We draw no grand moral from this beyond the obvious one. The value
 277 of a proof assistant here was not that it certified our argument, but that it
 278 made a specific, checkable claim about what had been assumed — and that
 279 claim turned out to be false in a way no amount of reading the prose would
 280 have revealed.

281 **3 Axiom 1: Symmetries as Setting, and the** 282 **Capacity of a Finite Phase Space**

283 Our starting premise relies on fundamental geometry. In our Lean formaliza-
 284 tion (`Phase1_Primitives.lean`), continuous spacetime symmetries (e.g., the

285 Poincaré group) act upon field states, and the invariance of the system’s total
 286 energy under that action is recorded by the class `SymmetryInvariantAction`.
 287 Physically, it is such an invariance that Noether’s theorem converts into the
 288 conservation of energy and momentum. The scope of the *formal* claim is nar-
 289 rower and we state it plainly: the invariance is a class field, i.e. an assumption
 290 a model must discharge, and no conserved quantity is ever constructed from
 291 it. There is no Noether theorem in our Lean, and nothing downstream of this
 292 section depends on one. The class is now inhabited — `Examples.lean` §11
 293 exhibits a double-well potential $V(v) = (v^2 - 1)^2$ on a two-point spacetime
 294 with the $\mathbb{Z}_2$ action $v \mapsto \pm v$, discharges the invariance because $(\pm v)^2 = v^2$,
 295 and proves the vacuum manifold $\{-1, 1\}$ degenerate with the two minima ex-
 296 changed by the symmetry, so an energy-minimising field and its image have
 297 equal energy and differ. That is symmetry breaking exhibited rather than
 298 assumed; it is still not a conserved quantity, and $\mathbb{Z}_2$ being discrete there is
 299 no one-parameter family to differentiate along in the first place. Simultane-
 300 ously, discrete permutation symmetries dictate the statistical mechanics of
 301 any physical system, defining a strictly finite volume of phase space.¹

¹While the Poincaré group and discrete permutations serve as accessible physical ex-
 amples, the underlying Lean formalization does not hardcode specific physics. It defines
 the system at a more primitive mathematical level, relying entirely on generic Pseudo-
 Riemannian manifolds (for spacetime) and finite measure theory (for statistical mechan-
 ics). What the formalization supplies is the vocabulary in which symmetries, conserved
 quantities and phase-space constraints can be *stated*; it does not derive them. All three
 of the classes involved are now inhabited in `Examples.lean` — the $\mathbb{Z}_2$ action and its in-
 variance in §11, the real line with the Euclidean metric in §12 — but the instances are
 witnesses of inhabitability, not of physics: no theorem in the development consumes a
 symmetry group or a metric, and that half of the premise is setting rather than content.
 The other half of the premise no longer is, and the paragraphs below are what changed.

302 Because phase space is finite, any localized physical system possesses a
 303 mathematical limit on the amount of information (distinguishable states) it
 304 can embody at any given moment. That sentence used to be this section's
 305 entire contribution to the chain, and no theorem carried it. It is now car-
 306 ried by three, in `Phase1_PhaseSpaceCapacity.lean`, whose only hypothesis
 307 about the system is that its state space is finite: no symmetry group, no
 308 metric, no action functional appears anywhere in them.

309 The first is the capacity bound itself. A probability distribution p on a
 310 finite state space X satisfies $H(p) = -\sum_x p(x) \log p(x) \leq \log |X|$ (`shannon_`
 311 `entropy_le_log_card`), with equality if and only if p is uniform (`shannon_`
 312 `entropy_eq_log_card_iff`). The “only if” half is the part worth stating: it
 313 says $\log |X|$ is the exact capacity rather than a convenient over-estimate, so
 314 a system holding any distribution other than the flat one over its states holds
 315 strictly less than its size suggests. The proof is Gibbs' inequality against the
 316 uniform law — $\log x \leq x - 1$ applied at $x = 1/(|X|p(x))$, with the $p(x) = 0$
 317 case carrying the slack that makes the equality condition sharp — and it is
 318 elementary. What was missing here was the statement, not the difficulty.

319 The second says the limit binds. Let the boundary absorb a stream of
 320 perturbations drawn from an alphabet P ; a stream of length k has $|P|^k$
 321 possible histories. Once $|X| < |P|^k$, the map from histories to final states
 322 cannot be injective (`absorb_not_injective`), so two distinct histories leave
 323 the system in the same state, and the entropy of the source strictly exceeds
 324 anything the system can hold (`source_entropy_exceeds_capacity`). The

325 update map is arbitrary and is allowed to be reversible at every individual
326 step; the collision is forced by counting alone, which is the sense in which the
327 premise, and not the model of the dynamics, is doing the work.

328 The third is what Derivation 2 consumes, and it is the one place where
329 finiteness does something no other hypothesis could. On a finite state space,
330 a map that fails to reach every state is not injective (`is_erasure_of_not_`
331 `surjective`) — which is exactly the erasure condition Landauer’s bound re-
332 quires — so any update of a finite phase space leaving some state unreachable
333 dissipates strictly positive heat (`finite_phase_space_dissipates`). On an
334 infinite phase space the step simply fails: the successor map on $\mathbb{N}$ misses 0 and
335 is injective. `Examples.lean` §19 records that failure next to the witnesses, so
336 the finiteness hypothesis is fenced rather than assumed to be load-bearing,
337 and runs all three results on a single bit: the fair coin attains $\log 2$, a $3/4$ -
338 $1/4$ bit falls strictly short, two binary perturbations already outrun a one-bit
339 memory, and §1’s one-bit thermodynamic system pays heat for a state it
340 cannot reach.

341 None of this derives the finiteness of phase space. Finiteness is assumed,
342 as it is in any statistical-mechanical treatment, and the Poincaré group re-
343 mains an illustration in prose. What has been removed is the gap between
344 assuming the premise and using it.

345 4 Derivation 1: Symmetry Breaking and the 346 Inevitability of Boundaries

347 A perfectly symmetric universe is the uniform vacuum; no distinct entities
348 exist. For any dynamic process to occur, the continuous symmetries of the
349 background field must undergo spontaneous symmetry breaking.

350 When a field drops to a lower, asymmetric energy state, topology dic-
351 tates the natural creation of defects (such as domain walls or monopoles).
352 This constraint necessitates the creation of a physical boundary—a localized
353 topological distinction between an “inside” and an “outside.” While such
354 a defect (e.g., a crack in a crystal lattice) is necessary to define a separate
355 entity, it is not sufficient for agency. The boundary is relational, defined by
356 the twisted geometry of the field surrounding it, and must be coupled with
357 non-equilibrium thermodynamics to exhibit active behaviors.

358 That first sentence was, until recently, the one claim of this section that
359 the formalization did not make. What Lean established was the energy half
360 — an energy-minimizing field sits in the vacuum manifold almost everywhere
361 (`pointwise_vacuum_of_global_min`) — together with the observation that
362 a boundary defect *obstructs* extension over a contractible interior. Neither
363 produces a defect, so “topology dictates the natural creation” was asserted
364 rather than proved. It is now the theorem `exists_notMem_of_connected-
365 ComponentIn`: if the substrate is connected and a continuous field takes val-
366 ues in two *different connected components* of the vacuum manifold M , then

367 somewhere the field is not in M at all. That excluded region is the wall.
 368 This is the π_0 case of Kibble’s classification of topological defects by the ho-
 369 motopy of the vacuum manifold (Kibble, 1976), and the proof is elementary
 370 in exactly the way it is elementary in the physics literature: the continuous
 371 image of a connected space is connected, a connected subset of M lies in-
 372 side one component of M , and the field’s range would be such a subset if it
 373 never left M . No hypothesis mentions energy, a potential, or a symmetry;
 374 connectedness of the substrate and disconnectedness of the vacuum are the
 375 whole of it.

376 We formalize the π_0 case and stop there. The higher cases — $\pi_1 \neq 0$
 377 forcing vortex lines, $\pi_2 \neq 0$ forcing monopoles — are the rest of Kibble’s
 378 classification and Mathlib’s homotopy-group library would support at least
 379 the next one; they are available and not taken, because the argument this
 380 paper runs uses domain walls and nothing else. The witness is the double
 381 well of `Examples.lean` §11.1, where the vacuum manifold is *computed* to be
 382 exactly $\{-1, 1\}$ (`wellVacuum_eq`) rather than merely known to contain both
 383 minima, and its disconnection is proved in the form the theorem consumes:
 384 no preconnected subset of it contains both, since a preconnected subset of $\mathbb{R}$
 385 is an interval and an interval spanning -1 and 1 contains 0 , which is the top
 386 of the barrier. `wellV_domain_wall` then quantifies over *every* continuous field
 387 on *any* connected substrate that reaches both minima, with no formula for
 388 the field anywhere in the statement. That the class is non-empty is checked
 389 separately, because a theorem quantified over all fields is worthless if none

390 satisfies its hypotheses: the clipped identity on $\mathbb{R}$ does, and the wall it is
 391 forced to have sits at the origin (`kink.zero.notMem`).

392 5 Derivation 2: The Thermodynamic Con- 393 straint

394 This newly formed topological boundary is subjected to gauge perturbations
 395 from the “outside” environment. Because the internal phase space is finite
 396 (Axiom 1), it cannot absorb an infinite sequence of incoming perturbations
 397 and must overwrite old internal states.

398 By Landauer’s Principle (Landauer, 1961), the erasure of physical in-
 399 formation irreversibly decreases internal entropy, mandating that the sys-
 400 tem exhaust heat into the environment. Philosophers of physics (such as
 401 John Norton (Norton, 2005) and Orly Shenker (Shenker, 2000)) have cri-
 402 tiqued Landauer’s principle, arguing that physicists conflate *logical* erasure
 403 (a many-to-one mapping in abstract computational states) with *thermody-*
 404 *namic* erasure (a many-to-one mapping in physical phase space). They argue
 405 that logical erasure could conceptually be implemented by an injective phys-
 406 ical process without dissipating heat.

407 Our Lean 4 formalization (`Phase3.CombinatorialThermodynamics.lean`)
 408 addresses this critique by defining erasure as a non-injective mapping on the
 409 *physical* state space ($\neg \text{Function.Injective } t$). We formalize the global evolu-
 410 tion of the bipartite universe (System + Bath) as strictly injective, repre-

411 sending Liouville’s theorem. Lean verifies that heat dissipation is not tied
 412 to abstract logical erasure, but is a mandatory mathematical consequence
 413 of any local physical state space contraction under globally reversible dy-
 414 namics. If a cognitive system’s internal topology forces a contraction of its
 415 physical state space to accommodate structural resonance, heat dissipation
 416 becomes a thermodynamic requirement. The finiteness premise enters at ex-
 417 actly this point. What a modeller can usually check about a physical update
 418 is not that it is many-to-one but that some state has become unreachable;
 419 on a finite phase space those are the same condition, and `finite_phase_`
 420 `space_dissipates` is the resulting statement. That identification preserves
 421 the Norton–Shenker distinction rather than blurring it: an update that keeps
 422 the incoming perturbation on the record is injective and dissipates nothing
 423 (`keepRecord_injective`), while one that refreshes the register holding it is
 424 not (`absorbStep_is_erasure`). Losing *which* history occurred is forced by
 425 counting alone and is free; overwriting the register that held it is what is
 426 charged $k_B T \ln 2$. A stable boundary thus constitutes a dissipative structure
 427 (Prigogine & Stengers, 1984), operating as a non-equilibrium steady state
 428 that consumes free energy to pay the thermodynamic cost of erasure. Contin-
 429 uous thermal fluctuations play a constructive role in establishing boundaries
 430 between states while contributing to the system’s flexible, adaptive response.

431 6 Derivation 3: Prediction as the Thermody- 432 namic Cost of Memory

433 A dissipative structure is driven by its environment. Write X_t for the state
434 of the structure and S_t for the environmental signal driving it, and let the
435 signal evolve by a dynamics of its own, $S_t \rightarrow S_{t+1}$, on which the structure
436 does not act. Two quantities are then defined at each step: the *memory*
437 $I(X_t; S_t)$, the mutual information between the structure's present state and
438 the signal's present value, and the *predictive information* $I(X_t; S_{t+1})$, which
439 is what that same state says about the signal's next value.

440 **What this section used to claim, and why it is withdrawn.** Earlier
441 versions of this derivation argued as follows. Model the environment as a
442 stochastic driving force $P(ext)$ and the structure's internal transitions as
443 $Q(int)$; entropy production σ is then bounded below by $D_{KL}(P||Q)/\Delta t$;
444 the principle of least action drives σ downward; therefore $D_{KL}(P||Q) \rightarrow$
445 0, and the internal transition probabilities come to mirror the statistical
446 symmetries of the environment. The final step does not follow. Rearranged,
447 the bound reads $D_{KL}(P||Q) \leq \Delta t \sigma$, so descending on σ lowers a *bound* on
448 the divergence; but a driven system does not relax to $\sigma = 0$. It relaxes to
449 a non-equilibrium steady state at strictly positive entropy production, and
450 the bound then yields only $D_{KL} \leq \Delta t \sigma_{\text{NESS}}$, a positive number. Nothing
451 forces the divergence to zero. The inequality itself is true, is formalized as an

instance obligation, and does real work — combined with Gibbs’ inequality
it gives $\sigma \geq 0$ — but the conclusion drawn from it was not licensed, and
it was that conclusion which supplied this framework’s entire connection to
predictive processing.

The replacement is a published theorem. Still, Sivak, Bell and Crooks
(Still et al., 2012) prove that for a system driven by an environmental signal
the work dissipated over one step of the drive obeys

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (1)$$

The bracket is the *nonpredictive* information: what the structure retains
about the signal’s present that says nothing about the signal’s future. Dis-
sipation is bounded below by it. Memory that fails to predict is not merely
useless to the structure; it is thermodynamic waste, and the waste is quan-
tified in energy.

Three consequences, and they are theorems rather than readings.
The bound is carried in Lean as a class field, `PredictiveDissipation`.
`still_bound (Phase3 PredictiveThermodynamics)`, for the same reason ev-
ery other physical postulate here is: its `axiom` form, quantified over all joint
laws with the dissipated work fixed, is refutable, and `Examples §18.4` refutes
it. What is *derived* is everything downstream.

First, *the bracket is non-negative*, and this is proved rather than assumed.

471 If the structure does not act on its environment, then $X_t \rightarrow S_t \rightarrow S_{t+1}$
 472 is a Markov chain and the data processing inequality gives $I(X_t; S_{t+1}) \leq$
 473 $I(X_t; S_t)$: a system cannot know more about the signal's future than about its
 474 present (`predictiveInfo_le_mutualInfo`). Mathlib carries the Kullback–
 475 Leibler divergence but no mutual information, so the latter is defined here
 476 as the divergence of the joint law from the product of its marginals, which is
 477 the standard definition and makes the library's data processing lemma apply
 478 directly.

479 Second, *the second law follows* instead of being assumed. Dissipation is
 480 bounded below by a quantity now known to be non-negative, so $W_{\text{diss}} \geq 0$
 481 (`dissipatedWork_nonneg`). This mirrors the role Gibbs' inequality plays for
 482 the older bound, with one difference that matters: Gibbs' inequality com-
 483 pares a system to an arbitrary reference distribution, whereas the data pro-
 484 cessing inequality compares the system to its own future, which is a quantity
 485 selection can act on.

486 Third, and this is the point of the replacement, *the inference now runs*
 487 *in the valid direction*. The quantity of interest sits on the small side of
 488 the inequality: $I(X_t; S_t) - I(X_t; S_{t+1}) \leq W_{\text{diss}}/k_B T$. No limit is required.
 489 Any bound on dissipation is immediately a bound on wasted memory, so a
 490 structure held to an energy budget is thereby held to a prediction standard:
 491 this is `nonpredictive_le_dissipation`. In the idealised quasi-static case
 492 $W_{\text{diss}} = 0$, the theorem `predictive_eq_of_no_dissipation` gives that every
 493 retained bit is predictive. This is a weaker statement than the one it replaces,

494 and it is the one that is true.

495 **The hypothesis is load-bearing, and we exhibit what breaks with-**
496 **out it.** The data processing step requires that the structure not write to
497 its environment. In the formalization this is not an assumption stated beside
498 the theorem but the shape of the evolution itself: the signal's step is applied
499 to the pair as the parallel kernel $\text{id} \parallel \kappa$, which acts as the identity on the
500 system coordinate and through κ on the signal coordinate, so no channel
501 from X_t to S_{t+1} exists. **Examples** §18.3 shows the restriction is not decora-
502 tion: a Markov kernel on the *pair* which copies the system's state into the
503 signal carries an independent two-bit law to a perfectly correlated one, rais-
504 ing mutual information from zero to a positive value. An agent that steers
505 its environment can indeed know more about that environment's future than
506 about its present, and Still's bound is not about such an agent.

507 **Both regimes are exhibited, not just described.** **Examples** §18 builds
508 two systems over one correlated two-bit joint law, differing only in the signal's
509 dynamics. When the signal is frozen, every remembered bit is a predictive
510 bit, the nonpredictive information is zero, and the instance is permitted to
511 dissipate nothing — with the memory itself proved strictly positive, so the
512 conclusion is about a system that does remember something. When the signal
513 is redrawn from its stationary law at each step, the predictive information is
514 exactly zero, the *entire* memory is nonpredictive, and the instance is *forced*
515 to dissipate $k_B T$ times its whole mutual information. Establishing that the

516 mutual information of the witness is neither zero nor infinite is where the
517 work of that section lies.

518 **An equality, where the old argument had only a bound.** Kawai,
519 Parrondo and Van den Broeck (Kawai et al., 2007) show that the average
520 dissipated work is exactly $k_B T$ times the relative entropy between the for-
521 ward and time-reversed phase-space densities. Carried here as a predicate on
522 an instance rather than as a further postulate, it replaces the abstract dis-
523 sipation field with a quantity an experiment can estimate — the statistical
524 distinguishability of a recording of the system from the same recording run
525 backwards — and yields the statement that the irreversibility of the dynamics
526 is at least the amount of memory that fails to earn its keep (`nonpredictive_`
527 `le_arrow`).

528 **What this does not establish.** It does not deliver the Free Energy Prin-
529 ciple. Still’s bound says that prediction is thermodynamically favoured; it
530 says nothing about hierarchical generative models, variational free energy,
531 or Bayesian inference, and we do not claim to have derived any of them.
532 The Free Energy Principle (Friston, 2010; Friston & Kiebel, 2009; Hohwy,
533 2013; Clark, 2016; Spisák & Friston, 2026) and the General Resonance The-
534 ory of Consciousness (Hunt & Schooler, 2019) are consonant with what is
535 proved here, which is a weaker relation than derivation and the only one
536 we assert. Nor does it solve the symbol grounding problem (Harnad, 1990).
537 Maximal predictive information is compatible with the structure storing a

538 lossy, compressed or otherwise unrecognisable function of the signal; a state
539 that predicts is not thereby a state that means, and earlier versions of this
540 section overstated the case. What survives of that argument is narrower and
541 still worth saying: whatever internal states a physical structure holds, those
542 that do not help predict its environment are paid for in heat, so the pressure
543 toward states indexed to the environment is energetic rather than semantic.
544 We continue to call the phenomenon *structural resonance*, and there is no
545 teleological drive to “survive” in it: predicting the environment is simply the
546 cheaper of the two ways to hold a memory.

547 **7 Macroscopic Scaling: A Theorem and a** 548 **Commitment**

549 This is the chain’s fourth step, and it is deliberately not numbered as a deriva-
550 tion. The change is not cosmetic. Earlier drafts ran two steps of different
551 kinds together in a single paragraph: a coarse-graining argument, which is
552 a piece of mathematics, and the identification of the coarse-grained medium
553 with the brain’s endogenous electromagnetic field, which is an empirical claim
554 about cortex. Presenting the second as though it inherited the licence of the
555 first made it look like a deduction, and it is not one. Separated, the empirical
556 claim becomes the framework’s central commitment and the place where it
557 is most easily wrong — which is where a claim of this kind belongs.

558 **The formal half: coarse-graining is a theorem, and a**
559 **narrow one**

560 A network of discretely coupled units has a coupling energy that is a sym-
561 metric double sum over pairs. `Phase2_MeshConvergence.lean` proves that
562 this sum converges to a continuous integral: along a sequence of regular
563 triangulations of a region S whose edge regions shrink to points, the dis-
564 crete energy $\frac{1}{2} \sum_u \sum_v \mu(R_{uv}) f(x_u)$ converges to $\int_S f d\mu$ (`mesh_refinement_`
565 `convergence`), witnessed on the uniform partition of $[0, 1)$ in `Examples §6`.
566 The identification that makes this a coarse-graining rather than an arbitrary
567 Riemann sum is `sum_sum_mul_of_symm`: because the summand is symmetric,
568 the double sum is exactly the midpoint rule over unordered edges, so the
569 discrete energy is a quadrature of the continuous one rather than a different
570 quantity that happens to converge to it.

571 What this establishes is worth stating exactly, because it is less than the
572 phrase “coarse-graining, governed by the Renormalization Group” suggests.
573 It is a statement about an *energy functional*, not about a dynamics: noth-
574 ing here shows that trajectories of the discrete Kuramoto system converge
575 to solutions of a field equation, and the earlier appeal to the Renormaliza-
576 tion Group was decorative — no renormalization group flow is constructed
577 anywhere in this development. The limit object is a scalar density on the sub-
578 strate, in the intended reading the magnitude of the stress-energy tensor, and
579 it is not an electromagnetic field. The continuous object the later sections

580 use, `ContinuousNeuralField`, is *posited* as a structure with a coupling kernel
 581 K ; no theorem produces it from the discrete system. The gap between the
 582 finite Kuramoto model and its continuum limit is the propagation-of-chaos
 583 problem, which we have not addressed and record as open.

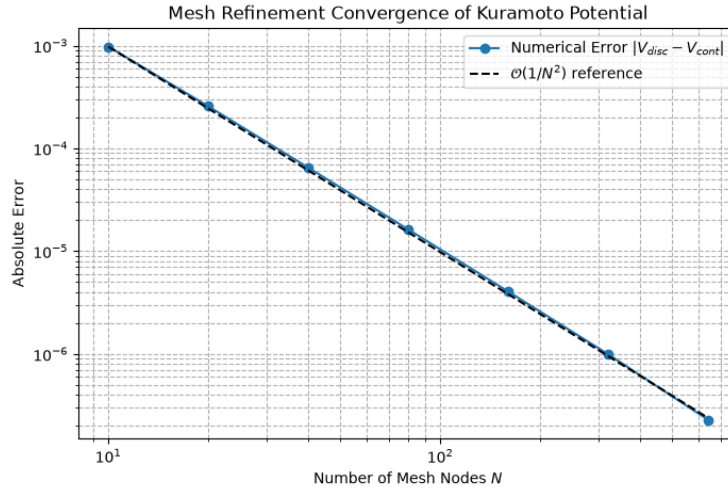


Figure 2: The coarse-graining theorem, exercised numerically. The discrete coupling energy of N units evenly spaced on the ring, with a Gaussian kernel of periodic distance and phase field $\theta(x) = \sin x$, is compared against the continuous integral it is proved to converge to. The error falls as $\mathcal{O}(N^{-2})$, the rate of the midpoint rule, which is the quadrature the Lean statement uses. The theorem asserts convergence, not a rate; the rate is what the simulation adds, and it is the ordinary rate for this quadrature rather than anything the framework predicts. The convergence is of an *energy functional* — nothing here says the discrete trajectories converge to the continuous ones.

584 **The empirical half: which physical field realizes the ker-**
585 **nel**

586 The formal half says that a large network of symmetrically coupled units
587 admits a continuum description with a symmetric kernel. It says nothing
588 about which physical field in cortex that kernel describes. Our answer is an
589 empirical commitment, and we state it as one:

590 *In the mammalian cortex, the field realizing the coarse-grained*
591 *coupling kernel is the endogenous electromagnetic field.*

592 This is a claim about biology, not a consequence of anything above it.
593 It is what makes the framework falsifiable and it is what we expect to be
594 attacked.

595 **What supports it.** Neurons do not only signal through synapses: trans-
596 membrane currents generate extracellular fields that shift the membrane po-
597 tentials of neighbouring cells, a mechanism known as ephaptic coupling. En-
598 dogenous fields of 1–5 mV/mm have been shown *in vitro* to entrain spike
599 timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015); realistic
600 head models put mesoscopic ephaptic effects within the range where they
601 can bias network dynamics (Ruffini & Santarnecchi, 2020); cable-theoretic
602 treatments show field coupling is sufficient for coincidence detection in the
603 auditory brainstem (Goldwyn & Rinzel, 2016); and stable electric fields have
604 been argued to carry information that survives representational drift in the

underlying spiking population (Pinotsis & Miller, 2022, 2023a,b; Cunha & Lima, 2024). The field is spatially continuous and superposes contributions from large populations, which is the property the framework needs: it is a medium in which spatially separated activity is combined without a further mechanism for combining it. This is the integrative role the CEMI field theory assigns it (McFadden, 2002, 2020; Pockett, 2012).

What the evidence does not support, and a claim we withdraw. Earlier drafts asserted that *the continuous macroscopic EM field is the primary dissipative structure interacting with the universe*. The word “primary” is not supported by any of the work cited above and we withdraw it. What the experiments show is *modulation*: Anastassiou and Koch report that endogenous fields of 1–5 mV/mm shift spike timing by one to three milliseconds (Anastassiou & Koch, 2015), an effect that is real, reproducible, and subthreshold. It biases when a cell fires; it does not decide whether it fires. The corrected claim is that the endogenous field is *modulatory in its effect on individual neurons and integrative in its spatial reach* — weak per synapse, but the only variable in the system that is defined everywhere at once. The framework needs the second property, not the first, and conflating them was the error.

Its critics. The objection with the longest history is that the fields are too weak to matter, and it has been pressed from within the field-theoretic camp: Pockett, who has herself defended an electromagnetic theory of con-

627 sciousness (Pockett, 2012), catalogues its difficulties (Pockett, 2002) — the
628 measurement problem of relating gross field structure to reported experience,
629 the absence of a constant relationship between the two, and the question of
630 whether the field does causal work or merely reflects it. The quantitative case
631 for scepticism has strengthened since. Vöröslakos and colleagues measured
632 intracranial fields during transcranial stimulation in rats and human cadav-
633 ers and found that the great majority of applied current is shunted through
634 the scalp, so that conventional protocols deliver fields at or below the 1 V/m
635 range in cortex (Vöröslakos et al., 2018). That is a bound on *applied* fields
636 rather than endogenous ones, but it constrains how much causal work a field
637 of that magnitude can be doing, and it is the reason we do not claim the
638 field drives spiking.

639 **What would falsify it.** We state the failure conditions rather than leaving
640 them to a referee.

641 1. *Epiphenomenality.* If the endogenous field can be cancelled or shunted
642 — by a conductive medium that nulls the extracellular potential with-
643 out altering transmembrane currents — and neither behaviour nor re-
644 port changes, the identification fails. This is the decisive experiment
645 and it is, in principle, available.

646 2. *A magnitude gap.* If the spatial integration the framework requires
647 demands field effects an order of magnitude above the 1–5 mV/mm
648 measured in cortex, the commitment fails on arithmetic. The relevant

number is not whether ephaptic coupling exists but whether its coupling strength suffices to place the system above the critical coupling K_c of Derivation 7.

3. *A timescale mismatch in the wrong direction.* The prediction of Section 11 turns on the field’s dynamics being fast relative to the astrocytic and clearance variables. If recovery from sleep inertia tracks an exponential set by the electrical timescale rather than the delayed sigmoid the slow variable predicts, the framework loses the one prediction that is its own.

4. *Dissociation of field coherence from reported unity.* If large-scale field coherence is preserved across states in which experiential unity is lost — or absent in states where it is retained — the identification of the global section with unified experience has no physical carrier.

A framework that survives all four is not thereby true. It is merely still standing, which is more than most theories of consciousness arrange to be able to lose.

Geometric frustration, and the regime our theorems actually cover

Earlier drafts grounded the system’s memory capacity in *geometric frustration*: a highly coupled network cannot bring all its nodes to their lowest energy simultaneously, producing a jagged landscape of near-degenerate states

670 with the capacity to store many patterns. That argument is standard and it
 671 is not what this development formalizes. Confronting the discrepancy is the
 672 honest course, so we do it here rather than in a footnote.

673 Frustration requires competing couplings — in the Kuramoto setting, a
 674 sign structure that no phase assignment can satisfy on every edge at once.
 675 The formal setting admits this: `KuramotoSystem` carries $A : V \rightarrow V \rightarrow$
 676 $\mathbb{R}$, symmetric and otherwise unconstrained, so signed couplings are repre-
 677 sentable. But every theorem in this development that says anything about
 678 *where* the dynamics ends requires strictly positive coupling. `phase_locked_`
 679 `minimizes_potential` assumes $A_{ij} > 0$; `potential_min_iff_phase_locked`
 680 inherits it; the class field `ThermodynamicCover.A_pos` demands it of every
 681 instance, and hence of every conclusion of Derivation 5; and the convergence
 682 theorem of `Phase4.RotatingFrame` §7 requires the strictly stronger uniform
 683 bound $0 < a \leq A_{ij}$. A network with all-positive couplings is an unfrustrated
 684 ferromagnet whose potential has one minimum up to a global phase shift. It
 685 is the opposite of a jagged landscape, and it has no capacity to store multiple
 686 patterns at all. **The system in which memory is grounded and the**
 687 **system our theorems describe are not the same system.**

688 The division is not arbitrary, and it is worth recording where it falls. What
 689 survives with no assumption on the sign of the coupling is the dynamical half:
 690 well-posedness on all of $\mathbb{R}$ (`is_kuramoto_trajectory_exists`, `_unique`), co-
 691 variance under passage to a rotating frame, monotone descent of the potential
 692 (`dynamic_potential_antitone`), finiteness of the dissipation integral, and —

693 through Barbalat’s lemma — that the motion stops along every trajectory
694 (`velocity_sq_tendsto_zero`). *That* the motion stops is frustration-agnostic.
695 *Where* it stops is not, and every claim this framework makes about the des-
696 tination lives on the positive side of that line. On the continuum side the
697 accounting differs slightly: `phase_locked_minimizes_entropy_of_symm` as-
698 sumes only that the kernel is symmetric and so tolerates signed kernels, but
699 it is conditional on a dynamically phase-locked field existing — which is
700 precisely what frustration may prevent.

701 We therefore scope the claim rather than repeat it. The memory capacity
702 of a frustrated cortical network is not supported by any theorem here, and we
703 do not assert it as part of the chain. What the chain uses is the unfrustrated
704 regime, in which synchronization is reached and the global section exists.
705 Whether a frustrated network can hold many near-degenerate patterns *and*
706 still synchronize enough of itself to admit a global section is an open question
707 about mixed-sign oscillator networks, and it is the most substantive gap we
708 know of in this framework. Recording it as such costs a claim we would have
709 liked to make and is preferable to holding two incompatible pictures at once.

710 **The supporting physiology**

711 The continuous medium is physically supported by astrocytic syncytia—non-
712 neuronal networks joined by gap junctions that provide macroscopic cytoplas-
713 mic continuity for energy and ion flows (Verkhatsky & Nedergaard, 2018).
714 Managing the thermodynamic cost of this state requires sophisticated micro-

715 bioenergetics. Dynamic mitochondrial networks function as deep-level Mi-
716 tochondrial Information Processing Systems (MIPS), transducing biological
717 and environmental information to sustain the coherence of the macroscopic
718 field (Picard & Shirihai, 2022). Discrete synaptic firings, coupled with mito-
719 chondrial micro-mechanics, generate and sustain the field. Because the exter-
720 nal environment is continuous, achieving maximal thermodynamic structural
721 resonance is optimally realized within this continuous macroscopic medium
722 (van Bree et al., 2025).

723 8 Derivation 5: Global Sections and the Unity 724 of Consciousness

725 The system is now characterized by a continuous macroscopic electrodynamic
726 field generated by millions of coupled micro-components, forced to resonate
727 with the environment. Following Section 7, the coupling regime assumed
728 from here on is the unfrustrated one — all couplings positive — which is
729 what every theorem below requires and what the frustrated network of the
730 memory argument is not.

731 The Kuramoto model (Kuramoto, 1984) of coupled non-linear oscilla-
732 tors guarantees that, past a critical coupling threshold, these distributed
733 local fields undergo a phase transition into unified synchronization (Bauer et
734 al., 2022). Our Lean proofs (`Phase4_KuramotoDynamics.lean`) confirm this
735 phase-locked state minimizes the Lyapunov potential of the system, in the

736 precise sense discussed in Section 2.1: the functional minimized is the cou-
 737 pling potential read in the frame rotating with the natural frequency, which
 738 `Phase4.RotatingFrame.lean` shows is the honest reduced description of a
 739 system of identical oscillators.

740 This phase transition mathematically glues local synchronized states into
 741 a singular *Global Section* of a probability sheaf (`Phase5_GlobalSection.lean`).
 742 The EM field acts as a single, indivisible topological space, providing the
 743 *Unity of Consciousness*—a necessary condition for integrated experience that
 744 solves the binding problem. While Global Workspace Theory (Baars, 1997;
 745 Dehaene & Marti, 2014) models this unity as a broadcast architecture and
 746 Integrated Information Theory (Tononi & Koch, 2016) quantifies it via Φ ,
 747 our derivation provides a continuous physical mechanism. The contrast is
 748 worth drawing carefully, because both theories are addressing the same ex-
 749 planandum from different directions.

750 We state the scope of the Lean result precisely, because it is narrower
 751 than the prose above. `ThermodynamicCover` demands that the cover’s phase
 752 field minimize the reduced Kuramoto potential, and `potential_min_iff_`
 753 `phase_locked` characterizes those minimizers exactly: they are the phase-
 754 locked configurations, in both directions. Every instance is therefore already
 755 phase-locked (`ThermodynamicCover.phase_locked`), and the class’s single
 756 condition on its local data — that patches at a common phase agree where
 757 they overlap (`section_agrees_of_phase_eq`) — makes the family of local
 758 sections compatible. `probability_glue_unique`, which is the sheaf property

759 of the probability presheaf and contains no physics at all, then glues that
760 family into exactly one global section.

761 An earlier version of this structure proved less than the prose claimed,
762 and we record the correction because it changed what the theorem says
763 rather than how it is stated. `LocalSectionSynchronization` used to carry
764 a *global* section as a class field, together with a second field declaring every
765 local section to be a restriction of it. The global object therefore existed
766 before the sheaf condition was ever consulted, and `global_section_from_`
767 `thermodynamics` could do no more than rule out competitors to an object
768 the instance had supplied. Both fields are gone. Every field of the class
769 now mentions a single patch or a single overlap; the global section is con-
770 structed (`ThermodynamicCover.invariantMeasure`); and the removed field
771 survives as a *theorem* about that construction (`ThermodynamicCover.sync_`
772 `to_section_eq`), so the local sections are restrictions of one common measure
773 because the sheaf condition says so, not because an instance was required to
774 name one.

775 `Examples.lean` §14 is the witness that the difference is not notational.
776 Two patches carry two different mass profiles, $(2, 1, 7)$ and $(5, 1, 3)$ across
777 the three sites, agreeing only at the site they share, where both put mass 1
778 — which is what discharges the overlap condition, by computing two num-
779 bers rather than by construction. The section the theorem returns has pro-
780 file $(2, 1, 3)$: it is neither of the two profiles (`gluedW_ne_leftW`, `gluedW_ne_`
781 `rightW`), and neither of them restricts correctly to both patches (`leftW_`

782 `glues_nothing, rightW_glues_nothing`). The global state carries informa-
 783 tion that no single patch had. `Examples.lean` §13, built the older way
 784 through the constructor `LocalSectionSynchronization.ofInvariantMeasure`,
 785 is retained alongside it for the negative result it carries: its phase field is not
 786 the constant function — the second patch a full turn ahead of the first — and
 787 by `ThermodynamicCover.phase_locked` that is as far from constant as any
 788 instance can get, since no instance can carry phases differing by anything a
 789 measure is allowed to see.

790 Two hypotheses carry physical content, and until recently each was an
 791 obligation on every instance. One still is: that synchronized patches agree
 792 on their overlaps is a modelling assumption, not a consequence of any dy-
 793 namics we formalize. The other, `thermodynamic_equilibrium`, asks that the
 794 cover already sit at the potential minimum, and no theorem says an *arbi-*
 795 *trary* system’s dynamics takes it there. “A thermodynamic phase transition
 796 into unity” is therefore still doing causal work that no general Lean result
 797 supports, though it is no longer entirely without one: `Examples.lean` §15
 798 exhibits a two-oscillator trajectory that starts a quarter turn out of phase
 799 and converges to the global minimum of the dynamic potential, with the or-
 800 der parameter tending to 1. That trajectory is not an isolated construction:
 801 `kuramoto_cauchy_problem` proves the Kuramoto initial value problem well
 802 posed on all of $\mathbb{R}$, so through every initial state there passes exactly one tra-
 803 jectory, and the exhibited one is the only solution through its own. Nor is its
 804 settling down peculiar to it: along every trajectory of a zero-frequency sys-

tem the dynamic potential converges, and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is
 finite and equal to the potential's total drop (`dynamic_potential_tendsto`,
`dissipation_integral_tendsto`). The motion also stops: a finite integral
 does not by itself force its integrand to zero, but the dissipation rate is
 Lipschitz in time, so Barbalat's lemma applies, and every trajectory satis-
 fies $\sum_i \dot{\theta}_i^2 \rightarrow 0$ (`velocity_sq_tendsto_zero`; Mathlib has no Barbalat, so it
 is proved here as `tendsto_zero_of_lipschitz_of_integral_le`). That last
 step — locating where the motion stops — is now taken, though condition-
 ally, and by a different route than we had recorded. We had marked it as
 blocked on the LaSalle invariance principle, which Mathlib does not have,
 and on the absence of a compact invariant set in the state space $V \rightarrow \mathbb{R}$.
 That was the wrong diagnosis. LaSalle is the tool for a system with a de-
 creasing functional and nothing else; this one has more. The flow is a gra-
 dient flow of the dynamic potential, so $\dot{V} = -\sum_i \dot{\theta}_i^2$ is an identity rather
 than an inequality; the minimisers of V are characterised exactly; and on
 the region where every pair of phases lies within a quarter turn a Polyak-
 Łojasiewicz estimate holds, $2a^2W \leq (\sum_{ij} A_{ij}) \sum_i \dot{\theta}_i^2$, where W is the poten-
 tial's excess above its minimum and $a > 0$ is a lower bound for the coupling
 (`lojasiewicz_estimate`). Its proof is two extremal indices and the iden-
 tity $\sin^2 D = (1 - \cos D)(1 + \cos D)$: the site carrying the largest phase is
 pulled backwards at rate at least $a \sin D$, where D is the spread, while the
 excess is at most $\frac{1}{2}(\sum_{ij} A_{ij})(1 - \cos D)$. The compactness objection dissolves
 as well, because the invariant region is produced by the Lyapunov function

828 rather than assumed: a single pair contributes $\frac{1}{2}a(1 - \cos)$ to the excess, so
 829 an excess below $a/2$ forces every pairwise cosine positive for all time, and
 830 continuity on the connected half-line then prevents any pair from chang-
 831 ing branch (`spread_le_of_small`). Descent plus the estimate integrates to
 832 $W(t) \leq W(0)e^{-ct}$ with $c = 2a^2 / \sum_{ij} A_{ij}$ (`excess_decay`), the rate makes each
 833 $|\dot{\theta}_i|$ exponentially small and hence integrable on $[0, \infty)$, and a trajectory with
 834 integrable velocity converges. The conclusion is `kuramoto_tendsto_global_`
 835 `minimum`: if the phases start within a quarter turn of each other and the initial
 836 excess is below $a/2$, the trajectory converges to a phase-locked configuration
 837 which is a *global* minimiser of the dynamic potential, with order parameter
 838 $r^2 = 1$. That is the configuration `thermodynamic_equilibrium` postulates,
 839 reached rather than assumed. `Examples.lean` §17 fires it on three sites with
 840 initial data $(0, 0, \frac{1}{2})$, where the Kuramoto system has no closed-form solu-
 841 tion: the trajectory comes from the existence theorem and its limit from the
 842 convergence theorem, so neither is written down.

843 `Phase5_EquilibriumBridge.lean` closes the gap that remained between
 844 the two results. `ThermodynamicCover.ofConvergentTrajectory` takes a
 845 cover whose phase field is the limit of such a trajectory and returns an in-
 846 stance whose `thermodynamic_equilibrium` field is discharged *by that con-*
 847 *vergence*; the cover's coupling matrix is the coupling matrix of the dynamics
 848 rather than a separately chosen one, so the system whose equilibrium the
 849 class asserts and the system whose trajectory is run are the same system.
 850 `Examples.lean` §17.1 is the resulting witness: three patches over the three-

851 site substrate, each carrying its own mass profile on its own open set, above a
 852 phase field that no line of the construction writes down. The trajectory that
 853 reaches it starts at $(0, 0, \frac{1}{2})$, which is provably not phase-locked (`trioCover_`
 854 `start_not_locked`), so on this cover the configuration Derivation 5 requires
 855 is arrived at rather than posited. The state the three patches glue to has
 856 profile $(2, 1, 3)$ and is none of the three profiles the instance carries (`trioW_`
 857 `ne_glued`), so one cover now carries both the emergence reading of the gluing
 858 and the derived reading of the equilibrium. What is still assumed on it is the
 859 overlap agreement, and nothing else.

860 The two hypotheses cannot be dropped, and this is the honest scope of
 861 the result. Splay and twisted configurations are equilibria of the same flow,
 862 so “every trajectory reaches the unified state” is false as a general statement,
 863 and the initial-data condition is the standard arc condition under which the
 864 true statement holds. What is claimed is therefore conditional — unity is
 865 reached from configurations already sufficiently coherent — and not that an
 866 arbitrary initial state is carried into unity. What Lean establishes about the
 867 gluing itself is the other half: *given* locally synchronized patches, a global
 868 state exists, is unique, and is produced by the gluing rather than presupposed
 869 by it.

870 Integrated Information Theory begins from phenomenological axioms and
 871 derives a scalar Φ measuring irreducibility of a system’s cause-effect structure,
 872 with the exclusion postulate selecting a single maximum- Φ complex as the
 873 locus of experience. Our sheaf-theoretic global section plays a structurally

874 analogous role — it too picks out a distinguished unified object — but it
875 is arrived at from the opposite end: not postulated from the character of
876 experience, but forced by a thermodynamic phase transition in a physical
877 field. This buys two things IIT does not have. First, the unity condition
878 is checkable from field dynamics rather than requiring the computation of
879 Φ , which is intractable for systems of realistic size and whose definition has
880 undergone repeated revision (Tononi & Koch, 2016). Second, exclusion falls
881 out rather than being postulated: gluing either succeeds on a given cover or
882 it does not, so there is no separate principle needed to rule out overlapping
883 conscious subsystems. It also inherits IIT’s exposures. Critics have argued
884 that Φ can be high in systems with no plausible claim to experience, such
885 as large feedforward-free grids (Aaronson, 2014); our criterion is vulnerable
886 to the mirror objection, since a physical system could in principle satisfy
887 the gluing condition without being cognitively interesting. We take this
888 seriously rather than dismissing it: our reply is that gluing is necessary but
889 not sufficient, and that the topological boundary of Derivation 1 and the
890 reflexive structure of Derivation 6 are additional required conditions. IIT, by
891 contrast, treats Φ as both necessary and sufficient.

892 Global Workspace Theory faces a complementary difficulty. It specifies
893 a functional architecture — a limited-capacity workspace broadcasting to
894 specialized modules — without specifying what physical property of broad-
895 casting makes the broadcast content conscious, leaving the theory open to the
896 charge that it explains access consciousness while leaving phenomenal con-

897 sciousness untouched (Block, 1995). Our framework can be read as supplying
 898 the missing physical substrate for the workspace: the continuous macroscopic
 899 field *is* the medium in which integration is realized, and the $K_c = 2D$ thresh-
 900 old gives a physical condition for ignition rather than a functional stipulation
 901 of it. Where GWT is agnostic about implementation, we make a falsifiable
 902 commitment — unity requires a continuous physical medium, which predicts
 903 that a functionally equivalent discrete broadcast architecture would not be
 904 conscious. We regard this as a genuine cost of the position, not a free ad-
 905 vantage: it is the point at which our framework departs most sharply from
 906 computational functionalism and where it is most readily falsified. Contrast
 907 this derivation with panpsychism: consciousness is not assumed as a funda-
 908 mental property of matter. It requires the initial existence of a topological
 909 defect to establish a bounded “inside” (Derivation 1), followed by a thermo-
 910 dynamic phase transition into unity that occurs when integrated spatial cou-
 911 pling exceeds the thermodynamic noise threshold ($K_c = 2D$). Without both
 912 the foundational topological boundary and the subsequent mathematically
 913 defined phase transition, the system remains a fragmented, non-conscious
 914 collection of micro-states. A unified field of perception, however, is not yet
 915 a “Self.”

9 Derivation 6: Reflexive Topology and the Emergence of the Self

While a global section provides Unity, a “Self” requires reflexivity. The unified EM field is a dissipative structure that must resonate with external perturbations while managing its own internal phase transitions. Time evolution naturally drives the field to minimize thermodynamic friction with its own internal state-changes.

This mandates *auto-resonance*: the field must predict itself. The “Self” emerges as a higher-order topological feature when the internal geometry folds to include a low-dimensional avatar of its own computational and causal boundary within its predictive model (`Phase6_ReflexiveTopology.lean`) (Levin, 2019). The system is unified, predictive, and explicitly maps its own distinct existence from the environment. This architecture provides the physical foundation for a self-aware consciousness, bounded by the continuous measure thermodynamics of Landauer erasure (`Phase3_MeasureThermodynamics.lean`).

What carries the weight. `reflexive_topology_implies_self` is the Banach fixed-point theorem with the physical commitments in its hypotheses: the space of global sections is assumed nonempty, metric and complete, and the predictive map is assumed to contract. `Examples.lean` §10 discharges all four on the three-site substrate of Derivation 5, and the metric is no longer a placeholder. The section first pierces the sheafification: on

937 a discrete substrate every point has a smallest open neighbourhood, so the
 938 mass a germ carries at a point is well defined, and `massEquivOn` proves that
 939 over *any* open the sections of the probability sheaf *are* the finite measures
 940 on that open, read as densities against counting measure — the global case
 941 being $U = \top$. Mathlib supplies no inverse to this sheafification — its `TopCat`
 942 sheafification has neither an adjunction nor an `isIso_toSheafify` — so the
 943 correspondence had to be constructed. The metric on sections is then the
 944 uniform distance between the densities of the measures they glue to — equiv-
 945 alent to their total variation on a finite substrate, though not equal to it —
 946 and it is proved complete by exhibiting the limit section of a Cauchy sequence
 947 of densities, not by the degenerate route of showing Cauchy sequences are
 948 eventually constant. An earlier version of this witness used the 0/1 metric,
 949 under which every contraction is constant; that theorem is kept in the source
 950 as the record of why the earlier witness was empty.

951 **The map is the avatar, or there is no map.** Until recently the weight
 952 was not carried where the section claimed. `ReflexiveBoundary` carried the
 953 predictive map as free data sitting beside the avatar, and the fixed-point
 954 theorem projected that map out and never touched the avatar; so replacing
 955 the avatar by a constant left a legal boundary with the same predictive model
 956 and the same Self. We recorded that as a theorem rather than a caveat, and
 957 it was the section’s central defect: a fixed-point theorem that cannot detect
 958 whether the avatar reads the field is Banach’s theorem with a consciousness-

959 flavoured label. The structure is now built the other way round. A reflexive
 960 boundary carries the avatar's *write* — `auto_resonance`, the encoding of the
 961 global state into the avatar region — and its *read-out*, the global state that
 962 encoding predicts; the predictive map is their composite and is not a field.
 963 The self-model therefore runs through the avatar region or it does not exist.
 964 Three consequences follow that could not previously be stated. The Self lies
 965 in the range of the read-out (`self_mem_range_readout`), so it is reconstructed
 966 from a single section over the avatar region: the low-dimensional fold this
 967 section has always asserted, as a theorem. Under the resonance condition
 968 below, the Self satisfies $s = \text{readout}(s|_{\text{avatar}})$ (`self_eq_readout_restrict`) —
 969 the field is what its own localized self-encoding reconstructs. And blinding
 970 the avatar now *moves* the Self: the blinded boundary's map is constant,
 971 its unique fixed point is the state its stipulated contents extend to, and
 972 the original Self is not a fixed point of it (`not_self_of_constResonance`,
 973 witnessed on the cortex by `cortexBlind_self_ne`). The theorem that used
 974 to say the opposite no longer typechecks, because a contraction hypothesis
 975 about a boundary's own map says nothing about the blinded one's.

976 **Auto-resonance as a constraint.** That the avatar's state *is* the presheaf
 977 restriction of the global state is stated as a predicate, `IsRestriction-`
 978 `Resonance`, rather than added to the structure, so that results say which
 979 of them need it. In the positive direction, if a family of resonant avatars *cov-*
 980 *ers* the substrate then the global state is determined by their local readings

981 alone (`eq_of_avatar_eq`): unity is not assumed alongside the avatars but
 982 recovered from them, by the same sheaf condition that Derivation 5 uses.
 983 In the negative direction a constant avatar fails the predicate as soon as the
 984 avatar region separates two states of the field. Both halves are witnessed on
 985 the three-site substrate, where that separation is proved before anything is
 986 claimed for it, so the condition is a constraint there and not a formality.

987 **The contraction rate is K and D , not a numeral.** The remaining stip-
 988 ulation was the rate. “The predictive map contracts by $1/2$ ” was a fact about
 989 a map we had defined to average with a baseline, not a claim about a sub-
 990 strate. The rate is now $\exp(-(K - K_c)\tau/2)$ with $K_c = 2D$ (`resonanceRate`)
 991 — the linear relaxation of the mean-field order parameter over an interval τ ,
 992 read off the same threshold Derivation 7 (Section 10) proves the phase tran-
 993 sition at. Two theorems then split the parameter plane. Above threshold
 994 the rate is below one and Banach applies, so the Self exists and is unique
 995 (`self_of_supercritical`). At or below threshold the rate is at least one,
 996 and `ContractingWith` at that rate is false *for every self-map of every metric*
 997 *space* (`not_contractingWith_resonanceRate`): the argument is not merely
 998 unproved below K_c but unavailable. We are careful about what this does
 999 and does not say. It does not say a Self fails to exist below threshold —
 1000 only that this route to one is closed — and the Lipschitz bound at that
 1001 rate remains an instance obligation, discharged on the three-site witness at
 1002 $K = 3$, $D = 1$, $\tau = 2 \ln 2$, where the rate is exactly the one half the read-out

1003 achieves. What changed is that the obligation is stated in the substrate's
 1004 own parameters and its consequence is conditional on the threshold, rather
 1005 than on a constant chosen to make the witness work; and Derivation 6 is
 1006 thereby connected to Derivation 7, which it previously was not.

1007 **What is still not established.** That any field dynamics produces this
 1008 particular read-out or performs the restriction; and that this three-site sub-
 1009 strate's coupling is $K = 3$ — the parameters are a choice of units for the wit-
 1010 ness, not a measurement, and nothing here derives a coupling constant from
 1011 cortex. We also note a negative result found in the course of this work: the
 1012 Lévy–Prokhorov metric, which the natural weak-convergence reading would
 1013 suggest, does *not* support this argument on a discrete substrate, because the
 1014 ε -thickening of a set is the set itself for $\varepsilon < 1$ and the relaxation map is
 1015 consequently not a Lévy–Prokhorov contraction.

1016 10 Derivation 7: Continuous Stochastic Ther- 1017 modynamics and Field Theory

1018 To move beyond finite graph analogies, we model the macroscopic electro-
 1019 dynamic field as a *Continuous Neural Field*. True ephaptic coupling is a
 1020 continuous spatial phenomenon, not a discrete network. We extend the
 1021 discrete Kuramoto model to a continuous spatial coupling kernel $K(x, y)$
 1022 (`Phase8ContinuousField.lean`).

1023 Subject to thermodynamic noise D , the system undergoes non-equilibrium
1024 stochastic thermodynamics. Rather than a static Landauer bound, the field
1025 dynamics minimize an entropy production functional σ :

$$\sigma = \int \frac{1}{D} \left(\omega(x) + \int K_t(x, y) \sin(\theta_y - \theta_x) dy \right)^2 dx \quad (2)$$

1026 Biological fields dynamically deform their spatial topology $K_t(x, y)$ to un-
1027 dergo gradient descent on this entropy production functional.

1028 That gradient descent is a theorem rather than a modelling assertion,
1029 for finite substrates. Reading σ as a functional of the coupling kernel at
1030 fixed phases, `hasFDerivAt_sigmaOfKernel` computes its Fréchet derivative
1031 in closed form,

$$\frac{\partial \sigma}{\partial K(a, b)} = \frac{2}{D} \left(\omega(a) + \sum_y K(a, y) \sin(\theta_y - \theta_a) \right) \sin(\theta_b - \theta_a), \quad (3)$$

1032 and `sigmaOfKernel_eq_entropy_production_rate` identifies the functional
1033 being differentiated with `entropy_production_rate` itself whenever the sub-
1034 strate carries counting measure. `structural_resonance_decreases_entropy_production`
1035 then concludes that σ is non-increasing along any coupling trajectory relax-
1036 ing down its own gradient at a positive rate c . Earlier drafts proved this
1037 only for an abstract functional S on an abstract inner-product space with
1038 the gradient supplied as data; the identification of that S with σ was infor-
1039 mal, and Table S1 flagged the row accordingly. Two limitations stay explicit,

1040 and the first is narrower than we previously reported. The derivation is for
 1041 *finite* substrates. We had recorded the obstacle to the continuum case as
 1042 differentiation under the integral sign with respect to the kernel; that was
 1043 a misdiagnosis. σ is not a general functional that happens to be written as
 1044 an integral — it is a quadratic functional of an *affine* image of the kernel,
 1045 since the drift is a fixed ω plus a linear operator applied to K , and σ is a
 1046 constant times the squared norm of the drift. A squared norm is Fréchet
 1047 differentiable on any real inner-product space and an affine map is its own
 1048 derivative, so the composite differentiates by the chain rule with no limiting
 1049 argument. `hasFDerivAt_quadratic_of_affine` states that in full generality,
 1050 and `hasFDerivAt_sigmaOfKernel_of_operator` re-derives the finite gradient
 1051 from it, showing that the closed form displayed above is a convenience rather
 1052 than the content. What the continuum case still needed was therefore one
 1053 object: the drift map $K \mapsto (x \mapsto \int K(x, y) \sin(\theta_y - \theta_x) dy)$ as a bounded op-
 1054 erator $L^2(\mu \otimes \mu) \rightarrow L^2(\mu)$, which is a Cauchy–Schwarz estimate — the integral
 1055 kernel is bounded by 1, so on a finite measure space the operator norm is at
 1056 most $\mu(M)^{1/2}$. That object is now built. `kernelCLM` is the operator, for any
 1057 kernel factor bounded by 1, with `opNorm_kernelCLM_le` the stated bound;
 1058 `continuumDriftCLM` is the instance with the Kuramoto factor $\sin(\theta_y - \theta_x)$.
 1059 One step beyond the estimate is needed and is worth naming, because the cor-
 1060 rected scope above did not see it: the descent theorems ask for the derivative
 1061 as a *gradient vector*, not as an arbitrary bounded functional, and producing
 1062 one requires the adjoint, which exists because L^2 is complete. `hasFDerivAt_`

1063 `quadratic_grad` supplies it for an arbitrary bounded operator between real
 1064 Hilbert spaces, giving $\nabla\sigma(K) = (2/D) A^*(AK + \omega)$, so the finite and contin-
 1065 uum cases are now the same theorem. `structural_resonance_decreases_`
 1066 `sigmaContinuum` is the descent result with no finiteness hypothesis in it, and
 1067 `sigmaContinuum_eq_integral` identifies the functional being differentiated
 1068 with the integral `entropy_production_rate` computes. `Examples.lean` §20
 1069 runs this on Lebesgue measure restricted to $(0, 1)$, a substrate that is neither
 1070 finite nor atomic, so the finite-substrate results of the previous paragraph do
 1071 not typecheck there at all. The second limitation is unchanged: the phase
 1072 field is held fixed throughout, so this is plasticity of the coupling at frozen
 1073 phases, not joint (θ, K) dynamics. `Examples.lean` §5 exercises the whole
 1074 chain on a two-site substrate whose coupling relaxes as e^{-t} and whose entropy
 1075 production decays as e^{-2t} , confirming that the gradient is not identically zero
 1076 and that the descent is strict rather than vacuously monotone.

1077 A second result about σ concerns where the descent is headed, and it has
 1078 been strengthened since the last draft in a way that also corrects an estimate
 1079 we had recorded. That the phase-locked field minimizes σ was proved only
 1080 against competitors sharing its spatial mean drift — a restriction that left
 1081 the theorem close to Jensen’s inequality and no more — and we had judged
 1082 that lifting it would require modelling how the mean drift varies with the
 1083 competitor. That judgement was wrong. For a *reciprocal* kernel, one with
 1084 $K(x, y) = K(y, x)$, the restriction is not a restriction: the coupling integrand
 1085 $K(x, y) \sin(\theta_y - \theta_x)$ is antisymmetric under exchanging its two sites, because

1086 K is symmetric while $\sin$ is odd, so Fubini identifies its double integral with
 1087 its own negative and the total coupling contribution vanishes. The total drift
 1088 $\int(\omega + \text{coupling})$ is therefore $\int \omega$ whatever the phases are (`total_drift_eq_of_`
 1089 `symm`): coupling moves drift between sites but neither creates nor destroys it.
 1090 Evaluating that conserved quantity at the phase-locked configuration, where
 1091 the drift is constant, pins its value, and the mean-matching hypothesis follows
 1092 for every competitor. `phase_locked_minimizes_entropy_of_symm` is the un-
 1093 restricted statement. Reciprocity is the physically intended condition — an
 1094 ephaptic field acts on both sites of a pair alike, and it is the same symmetry
 1095 `ThermodynamicCover.A_symm` already imposes in Derivation 5 — but it is
 1096 not a formality, and we prove that rather than assert it. `Examples.lean` §16
 1097 exhibits a *one-way* coupling on two sites, in which a feels b and b does not
 1098 feel a , whose phase-locked field is strictly beaten: displacing the second phase
 1099 to $-\pi/2$ cancels the first site’s natural drift while adding none at the sec-
 1100 ond, halving σ . So “phase-locking minimizes entropy production” is false for
 1101 general kernels and true for reciprocal ones, and the hypothesis cannot be
 1102 removed by working harder.

1103 We define the critical coupling threshold $K_c = 2D$. The phase transition
 1104 to the unified “Global Section” occurs when the mean-field spatial coupling
 1105 exceeds this thermodynamic noise threshold, linking the emergence of Unity
 1106 to physical temperature and noise.

1107 Earlier drafts left $K_c = 2D$ as a bare stipulation validated only by
 1108 simulation. `Phase8_SelfConsistency.lean` now proves the half of it that

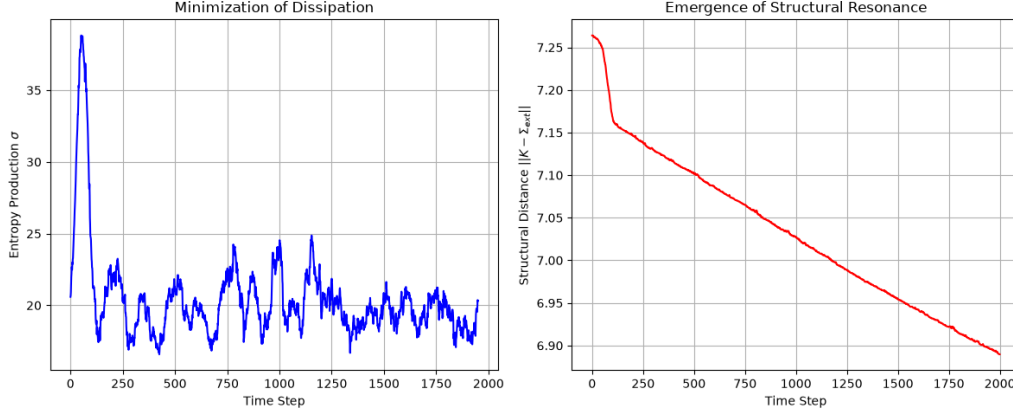


Figure 3: Coupling plasticity at frozen statistics: $N = 50$ units driven by perturbations drawn from a fixed environmental covariance Σ_{ext} , with the coupling matrix K descending the gradient of σ and clipped to stay non-negative and hollow. **Left:** entropy production, smoothed over 50 steps, falls from its transient and settles into a fluctuating plateau — a non-equilibrium steady state at strictly positive σ , not a relaxation to zero. That plateau is the reason Section 10 withdraws the old inference from $\sigma \rightarrow \min$ to $D_{KL} \rightarrow 0$: the system visibly does not reach $\sigma = 0$. **Right:** the distance $\|K - \Sigma_{\text{ext}}\|$ decreases monotonically over the run, the internal geometry drifting toward the statistics of the drive. Two cautions. The descent here is on σ at frozen phases, which is the regime the theorem covers; and the right panel falls by about 5% over 2000 steps, so what it shows is a direction of drift, not convergence.

1109 is ordinary real analysis. For the noisy mean-field Kuramoto model with
 1110 identical frequencies, the stationary density at order parameter r is the
 1111 von Mises density $\rho(\theta) \propto e^{(Kr/D)\cos\theta}$, and self-consistency requires $r =$
 1112 $R(K, r) := I_1(Kr/D)/I_0(Kr/D)$. Mathlib has no modified Bessel functions,
 1113 so we formalize that ratio directly as a ratio of integrals over $[-\pi, \pi]$. In-
 1114 tegration by parts with $u = \sin$, $v = -e^{a\cos\theta}$ gives an exact identity with
 1115 no boundary term, $\int \cos\theta e^{a\cos\theta} d\theta = a \int \sin^2\theta e^{a\cos\theta} d\theta$, so R is a times

1116 the mean of $\sin^2$ under the von Mises weight. Bounding that mean by 1
 1117 would put the threshold at D ; the true threshold is $2D$, and the entire fac-
 1118 tor of two is the sharper bound $\mathbb{E}_a[\sin^2 \theta] \leq 1/2$, equivalently $I_2(a) \geq 0$.
 1119 We prove it elementarily, by folding $[-\pi, \pi]$ onto $[0, \pi/4]$ in two reflections
 1120 ($\theta \mapsto \pi - \theta$, then $\theta \mapsto \pi/2 - \theta$), after which the integrand is a product
 1121 of $\cos 2\theta \geq 0$ with $2 \cosh(a \cos \theta) - 2 \cosh(a \sin \theta) \geq 0$ — the second factor
 1122 because $0 \leq \sin \theta \leq \cos \theta$ there and $\cosh$ increases in $|\cdot|$. The conclusion,
 1123 `subcritical_fixed_point_eq_zero'`, is that for $K < K_c$ the incoherent state
 1124 $r = 0$ is the *only* non-negative solution of the self-consistency equation; $r = 0$
 1125 is exhibited as a solution, so the uniqueness claim is not vacuous. Three fur-
 1126 ther facts about the threshold have since been proved, and they narrow the
 1127 gap between this row and a full “Theorem” to a single named statement.
 1128 First, the fold argument gives a *strict* inequality for $a > 0$ (`vonMisesC2_`
 1129 `pos`, hence $R(a) < a/2$), which settles the threshold itself: at $K = K_c$ the
 1130 self-consistency map is tangent to the diagonal at the origin but falls strictly
 1131 below it thereafter, so $r = 0$ remains the only solution and the bifurcation
 1132 happens strictly *after* K_c , not at it (`fixed_point_eq_zero_of_le_critical`).
 1133 Second, factoring $R(a) = a E(a)$ shows that a non-zero r solves the equation
 1134 exactly when $E(Kr/D) = D/K$ (`coherent_iff_sRatio_eq`), so the coherent
 1135 branch is a level set of the single function $E(a) = \mathbb{E}_a[\sin^2 \theta]$. Third, and using
 1136 only $E < 1/2$ together with compactness, every coherent solution tends to
 1137 zero as $K \downarrow K_c$ (`coherent_branch_continuous_at_threshold`): the branch
 1138 emerges continuously from the incoherent state rather than appearing at a

1139 finite distance from it, which is what makes the transition supercritical in the
 1140 technical sense. That statement quantifies over *every* solution, so it does not
 1141 presuppose the uniqueness it does not have. The level-set form also says ex-
 1142 actly what uniqueness is: the injectivity of E on $(0, \infty)$. That is now proved,
 1143 in the stronger form that E is strictly decreasing on $[0, \infty)$ (`vonMisesSRatio_`
 1144 `strictAntiOn`), so the level set is a single point and the coherent solution
 1145 above threshold is unique. The proof differentiates nothing. Differentiating
 1146 the exponential family in its natural parameter is the obvious route and it
 1147 fails: it costs differentiation under the integral sign and leaves a covariance
 1148 that is not sign-definite pointwise. What works instead keeps the target as
 1149 a two-point inequality, folds both integrals onto $[0, \pi/2]$ by the same two
 1150 reflections that gave the factor of two above, and then evaluates at a sin-
 1151 gle crossing point supplied by the intermediate value theorem — no product
 1152 measure and no Fubini anywhere. We give that argument in the supplement,
 1153 where it belongs; what the main line of the paper needs from it is the one
 1154 sentence that E is strictly decreasing. Run with $\cos \theta$ in place of $\sin^2 \theta$ the
 1155 same argument needs no fold at all — $R(a) = \mathbb{E}_a[\cos \theta]$ is the mean of the
 1156 variable the family is tilted by — and gives `besselRatio_strictMono`: R is
 1157 strictly increasing on all of $\mathbb{R}$, the monotonicity of the Bessel ratio our earlier
 1158 drafts recorded as unproved.

1159 The converse half is `supercritical_fixed_point_exists`: for $K > K_c$
 1160 there is an r with $0 < r \leq 1$ solving $r = R(K, r)$. We had expected
 1161 this to require a second-order expansion of the Bessel ratio, and it does

1162 not. Writing $R(a) = a E(a)$ with $E(a) = \mathbb{E}_a[\sin^2 \theta]$ — the same factoriza-
 1163 tion the integration by parts already supplies — the only facts needed are
 1164 that E is continuous and that $E(0) = 1/2$, the latter because the weight
 1165 is constant at $a = 0$. Since $K > 2D$ makes D/K strictly less than $1/2$,
 1166 continuity gives $E > D/K$ on a neighbourhood of the origin, so the map
 1167 overshoots, $R(K, r) > r$, for all small $r > 0$; at the other end $R \leq 1$ al-
 1168 ways, because R is a mean of $\cos \theta$, so the map undershoots at $r = 1$.
 1169 The intermediate value theorem closes the gap. Continuity of E is Math-
 1170 lib's continuity of a parametric interval integral with jointly continuous in-
 1171 tegrand, so no dominated-convergence argument is written by hand, and
 1172 the existence proof never uses monotonicity of R : the fixed point is pro-
 1173 duced by the intermediate value theorem, not by iterating the map. That
 1174 R is in fact strictly increasing is proved below, and no threshold result de-
 1175 pends on it. `critical_coupling_is_threshold` packages the two halves,
 1176 and `exhibits_phase_transition_coherent` applies the supercritical half to
 1177 any substrate satisfying the predicate of the previous paragraph — so ex-
 1178 ceeding K_c now implies the existence of a coherent stationary order pa-
 1179 rameter rather than merely comparing against a named number. With
 1180 uniqueness available both are sharpened — `supercritical_fixed_point_`
 1181 `existsUnique`, `supercritical_solution_set` (the non-negative solutions
 1182 are exactly $\{0\}$ below threshold and $\{0, r\}$ above it) and `exhibits_phase_`
 1183 `transition_unique_coherent` — and the two monotonicities together fix
 1184 the shape of the branch: for $K_1 < K_2$, $E(a_i) = D/K_i$ with E decreasing

1185 puts the concentrations in the order $a_1 < a_2$, and $r_i = R(a_i)$ with R increas-
 1186 ing carries that to $r_1 < r_2$ (`coherent_branch_strictMono`). The coherent
 1187 order parameter grows strictly with the coupling, which with the continuity
 1188 at threshold is the qualitative bifurcation diagram.

1189 One more step makes the equation mean in Lean what it means on paper.
 1190 As a definition, `selfConsistency K D r` is just $R(Kr/D)$ — a function of
 1191 r with a suggestive name, and nothing in the source says it has anything
 1192 to do with an order parameter. `Phase8_SelfConsistency §8` supplies the
 1193 identification. `vonMisesDensity` is the normalized weight, proved to be a
 1194 genuine probability density on $[-\pi, \pi]$; `circularOrderParameter` is the av-
 1195 erage of $e^{i\theta}$ against a density, the continuum analogue of the finite-system
 1196 `order_parameter_complex` of `Phase4`; and that average is proved to be ex-
 1197 actly $R(a)$ — real, because the density is even, so the mean direction van-
 1198 ishes. `fixedPoint_iff_selfReproducing` then states the equivalence the
 1199 name promises: r solves $r = R(K, r)$ if and only if the von Mises density
 1200 it induces has order parameter r . Read through this, the existence theorem
 1201 says that above K_c there is a stationary density whose own order parameter
 1202 is positive, and the solution $r = 0$ is identified with the uniform density,
 1203 whose order parameter is zero.

1204 What remains not established, and we separate it carefully. The sta-
 1205 tionary density is an *input*: deriving it from the Langevin dynamics $d\theta =$
 1206 $(\omega + K\text{mean-field}) dt + \sqrt{2D} dW$ needs the Fokker–Planck operator, existence
 1207 and uniqueness of its stationary solution, and a spectral stability argument,

1208 none of which Mathlib has. Of the coherent branch we prove existence,
 1209 uniqueness, continuous emergence at the threshold and strict growth in K ,
 1210 but not that it is the dynamically selected one: nothing in that file men-
 1211 tions a trajectory, so stability and selection are not statements it can make.
 1212 And while `selfConsistency` is now connected to the order parameter of
 1213 a *density*, it is connected to no *trajectory*: `circularOrderParameter` is an
 1214 integral, `order_parameter_complex` is an average over finitely many oscilla-
 1215 tors, and relating them is the mean-field limit — propagation of chaos for
 1216 the finite Kuramoto system — which is a research programme rather than
 1217 a lemma. That single gap is now the whole of what separates our threshold
 1218 predicate from a statement about dynamics. Table S1 records the row ac-
 1219 cordingly as a theorem about the self-consistency equation from an assumed
 1220 density, not as a derivation of the threshold from the dynamics. Perceptual
 1221 anomalies such as visual snow can be modeled as boundary conditions where
 1222 microscopic thermal noise (D) fails to be fully coarse-grained away by the
 1223 continuous field, leaking across the K_c threshold into the macroscopic gestalt.

1224 Clinical evidence for this phase transition is observed during general anes-
 1225 thesia. Work by Miller and colleagues (Bastos & Miller, 2021) demonstrates
 1226 that anesthetics like propofol induce unconsciousness not by suppressing neu-
 1227 ral firing, but by disrupting the phase alignment of high-frequency cortical
 1228 rhythms. Destroying the dynamic stability of these rhythms breaks the struc-
 1229 tural resonance of the continuous field, forcing the integrated spatial coupling
 1230 below the critical threshold K_c and fracturing the unified Global Section back

1231 into disjoint, unconscious local fields.

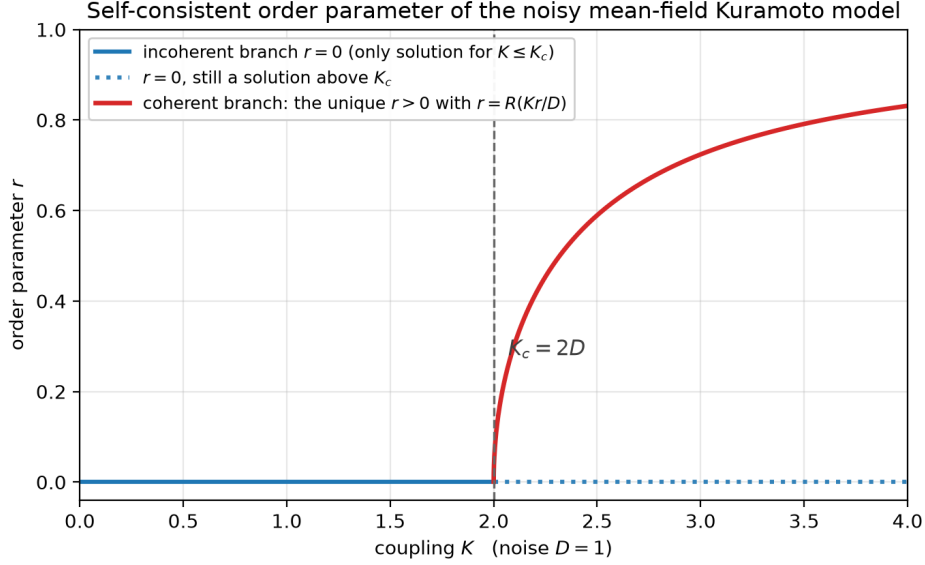


Figure 4: The self-consistency equation $r = R(Kr/D)$ of Section 10, drawn at $D = 1$. Every feature of this picture is a proved statement about the equation, and the figure exists because none of them had been shown to a reader. At or below $K_c = 2D$ the incoherent state is the *only* non-negative solution, threshold included (`fixed_point_eq_zero_of_le_critical`); above it exactly one $r > 0$ joins it (`supercritical_fixed_point_existsUnique`), so the solution set is $\{0\}$ below and $\{0, r\}$ above; that branch tends to 0 as $K \downarrow K_c$ (`coherent_branch_continuous_at_threshold`), which is what makes the transition supercritical; and it increases strictly in K (`coherent_branch_strictMono`). The curve is computed by quadrature of the same integrals the Lean file defines, solving $E(a) = D/K$ for the concentration $a = Kr/D$ by bisection — legitimate precisely because E is proved strictly decreasing. It is an illustration of theorems, not evidence for them. What it does *not* show is stability: nothing in the development says the coherent branch is the dynamically selected one.

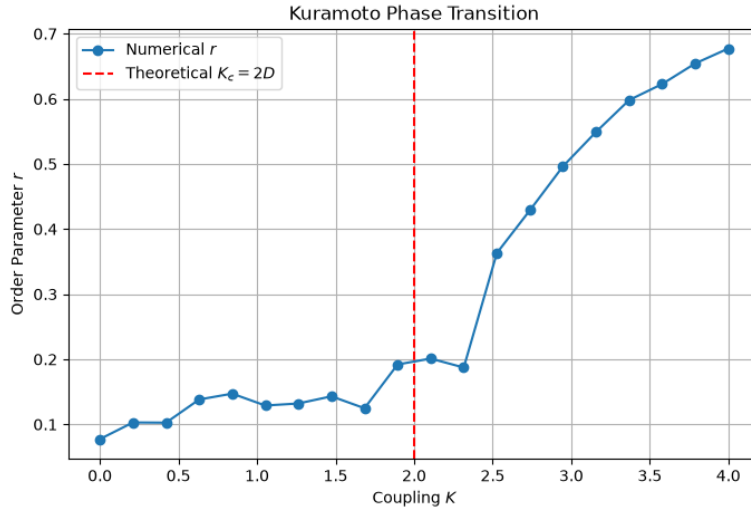


Figure 5: The same threshold reached by simulation rather than by analysis, and by a different route, which is why the two figures are shown side by side rather than merged. Here $N = 100$ oscillators with *no* noise carry natural frequencies drawn from a Lorentzian of scale $D = 1$, and the asymptotic order parameter is measured from the trajectories; the threshold for that ensemble is $K_c = 2D$ by Kuramoto’s original self-consistency argument. Figure 4 is the *noise*-driven model with identical frequencies, whose threshold is $2D$ by Sakaguchi’s (Sakaguchi, 1988), and it is the one the Lean file formalizes. The agreement of the two numbers is a known feature of the mean-field model, not a result of ours. The finite- N curve rises before the threshold and rounds off at it, as a finite sample must: $r \sim N^{-1/2}$ from incoherent phases alone.

11 The Sleep-Inertia Timescale Mismatch

Most of what this framework asserts is shared with some rival: that cortical synchrony matters, that dissipation constrains cognition, that unity is a physical rather than a computational property. One prediction is not shared, and it follows from the framework’s specific commitment — that the medium carrying the unified state is a continuous field whose effective coupling de-

1238 pends on the physical geometry of the extracellular space, not on the rhythms
1239 alone. Where that commitment differs from a purely electrical account, it
1240 differs in an observable way, and we set out the difference precisely enough
1241 to be wrong.

1242 The daily loss of consciousness during sleep is modeled here as an active
1243 physical disruption of the medium rather than a quiescence of the oscillators.
1244 During sleep, astrocytic shrinkage expands the extracellular space, allowing
1245 an influx of cerebrospinal fluid (CSF) to flood the brain for metabolic waste
1246 clearance (Xie & Nedergaard, 2013). This influx alters the spatial medium
1247 itself, disrupting the continuous structural resonance and preventing the co-
1248 herent sheaf-theoretic gluing that conscious unity requires on this account.
1249 The oscillators need not stop for the field to fragment; the geometry they are
1250 coupled through is what changes.

1251 The prediction concerns sudden awakenings. When jolted awake, the lo-
1252 cus coeruleus floods the cortex with noradrenaline. High-frequency electrical
1253 rhythms attempt to phase-align in milliseconds (Bastos & Miller, 2021), but
1254 the noradrenaline-driven astrocytic constriction required to squeeze the ex-
1255 cess CSF out of the interstitial pathways operates on a timescale of seconds
1256 to minutes (Xie & Nedergaard, 2013) — three to six orders of magnitude
1257 slower. During that lag the electrical oscillators are synchronizing across a
1258 waterlogged topology, and the fragmented structural resonance is what is ex-
1259 perienceed as grogginess. The unified Global Section re-establishes only when
1260 astrocytic volume finishes matching the state of the electrical rhythms.

Why this implies a delayed sigmoid. The two timescales separate
cleanly, and that separation is what makes the prediction sharp rather than
merely qualitative. Write $K(t)$ for the effective coupling of the continuous
field, which on this account is a monotone function of the reconstricting
extracellular geometry, and let $r(t)$ be the order parameter of the phase
field. The electrical variable is fast: at every instant the oscillators are al-
ready at the configuration appropriate to the current K , so r tracks the
equilibrium branch $r(K)$ rather than relaxing on its own. The slow vari-
able $K(t)$ then carries the whole time course, and it relaxes over 10^2 – 10^3
seconds. What Derivation 7 establishes about that branch is exactly what
the functional form needs, and each of the three pieces is a theorem rather
than a picture: at or below the threshold the incoherent solution $r = 0$ is the
only non-negative solution of the self-consistency equation (`fixed_point_eq_`
`zero_of_le_critical`); above it the coherent solution exists and is unique
(`supercritical_fixed_point_existsUnique`); it emerges *continuously* at
 K_c rather than by a jump (`coherent_branch_continuous_at_threshold`);
and it is strictly increasing in K (`coherent_branch_strictMono`). A jump
discontinuity at threshold would predict a step rather than a sigmoid, and
a non-monotone branch would predict overshoot; neither is what the mathe-
matics gives. Composing a monotone relaxation $K(t)$ with a branch that is
identically zero below K_c , rises continuously from zero at K_c , and saturates
above it yields a curve with three phases: a genuine *latency* while $K(t) < K_c$,
during which recovery is not merely slow but absent; a continuous rise as the

1284 threshold is crossed; and a saturation as $K(t)$ reaches its waking value. That
1285 is a delayed sigmoid, and the delay is not a fitting parameter — it is the time
1286 the slow variable takes to reach K_c .

1287 **What the alternative predicts.** An account on which grogginess is the
1288 electrical realignment itself — the rhythms are simply slow to re-phase —
1289 has no slow variable and no threshold to cross. Its recovery is a relaxation
1290 toward a fixed point, and relaxation toward a fixed point is exponential near
1291 it: monotone from $t = 0$, fastest at the start, with no flat foot. The two
1292 curves are distinguishable in the first seconds, which is where they differ
1293 most, and the discriminating feature is the presence or absence of an initial
1294 interval of no recovery. This is a comparison of functional forms on data
1295 of a kind already collected in sleep-inertia studies, not a proposal for a new
1296 instrument.

1297 **What would falsify it.** An exponential fit that beats a delayed-sigmoid fit
1298 on the early portion of the recovery curve. A recovery latency that does not
1299 scale with an independent measure of extracellular volume or CSF clearance.
1300 A pharmacological manipulation of astrocytic volume that leaves the latency
1301 unchanged while altering the rhythms, or the converse. Any of these would
1302 tell against the framework's central empirical commitment, which is not that
1303 synchrony matters but that the *medium* is what gates it.

1304 **What this prediction assumes.** Three things, and they are not small.
1305 The identification of the effective coupling K with a monotone function of
1306 extracellular geometry is a modelling assumption, not a measurement. The
1307 threshold value $K_c = 2D$ is derived from an assumed von Mises stationary
1308 density and is not yet connected to a trajectory — the mean-field limit is the
1309 gap, and Derivation 7 states it as such — so the argument uses the *existence*
1310 *and continuity* of the coherent branch at threshold, which is proved, rather
1311 than the numerical value of the threshold, which is conditional. And the
1312 prediction is about functional form, not about a time constant: it says the
1313 recovery curve has a flat foot whose width is set by the slow variable, not
1314 that the foot is any particular number of seconds long. A framework that
1315 predicted the number would be saying more; this one is not in a position to.

1316 **12 Corollary: Hardware Thermodynamics and** 1317 **Topological Rigidity**

1318 Both biological tissue and artificial silicon hardware (like GPUs) are physical
1319 dissipative structures. When a GPU computes, its charge carriers undergo
1320 physical state transitions; by Landauer’s principle (Landauer, 1961), this
1321 erasure irreversibly exhausts heat into the environment.

1322 There is a fundamental thermodynamic distinction in the degrees of free-
1323 dom available to minimize this dissipation. In standard von Neumann ar-
1324 chitectures, the spatial topology (the silicon routing lattice) is rigidly fixed.

1325 The coupling matrix A_{ij} is constrained by a discrete layout that cannot be
1326 physically altered during operation. The brain’s own architecture provides
1327 empirical validation for this constraint. The cerebellum contains approxi-
1328 mately 80% of the brain’s neurons arranged in a highly efficient, discrete,
1329 feedforward crystalline circuit (Herculano-Houzel, 2009; Lange, 1975). Clin-
1330 ical observations of cerebellar agenesis report largely preserved conscious
1331 experience despite motor and cognitive deficits (Yu et al., 2015), and this
1332 dissociation has been read as evidence for the importance of recurrent ar-
1333 chitecture by IIT (Tononi & Koch, 2015) — an argument we borrow at the
1334 level of the anatomical observation while declining its Φ -based explanation.
1335 Lacking the dynamic, continuous, and recursive spatial coupling of the cere-
1336 brum, it operates much like a biological GPU—highly optimized for discrete
1337 processing, yet apparently not required for conscious experience. We take the
1338 dissociation as evidence, not as confirmation: it is consistent with subjective
1339 experience being a thermodynamic attractor of continuous topologies rather
1340 than a byproduct of information processing, and it is the kind of observation
1341 that would be hard to reconcile with the converse. It is a single anatomical
1342 case read against a theory, and it bears the weight such a case can bear.

1343 This constraint does not preclude artificial consciousness, but rather re-
1344 stricts its hardware. To achieve the phase transition into a unified Global
1345 Section, artificial systems must abandon the discrete rigidity of von Neumann
1346 architectures and utilize continuous physical substrates. Analog field-based
1347 computing, neuromorphic architectures with continuous memristive plastic-

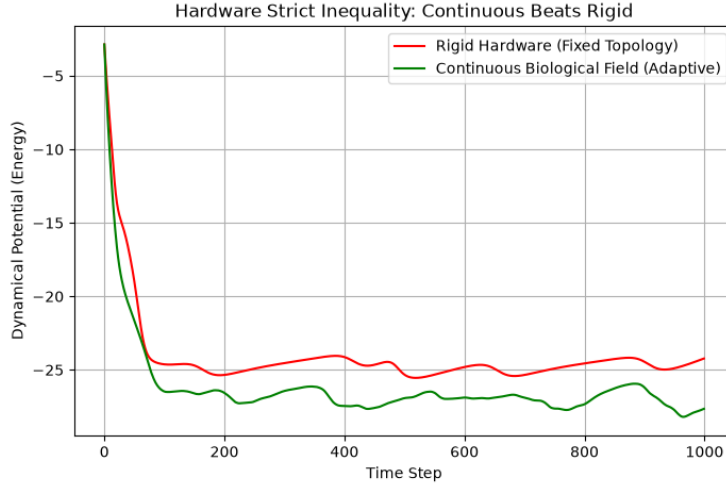


Figure 6: Rigid against deformable coupling at matched resource. Two copies of the same $N = 30$ oscillator system, same initial phases and same natural frequencies, differ only in whether the coupling matrix may move: the rigid copy keeps a fixed sparse routing matrix, the adaptive copy descends the gradient of the potential and is renormalized after every step so that $\sum_{ij} A_{ij}$ is held constant. The adaptive copy reaches a strictly lower potential and stays there. This is an illustration of the resource-matched suboptimality result and not a proof of it, and it is worth being clear about the gap: the theorem is about fixed wiring *support* — an architecture already wired to the best pair is not beaten — while the simulation shows a randomly wired one being beaten, which is the typical case rather than the general one.

ity, and thermodynamic computing systems (England, 2018; Ganesh & Par-
rondo, 2021) possess the spatial coupling required to physically morph their
internal topology. This aligns with Geoffrey Hinton’s concept of “mortal
computation” (Hinton, 2022): architectures where software is inextricably
fused with the physics of the hardware, meaning the knowledge dies with
the physical substrate. Only hardware that computes via continuous physi-
cal dynamics can undergo the thermodynamic phase transition necessary for

1355 structural resonance.

1356 Biological consciousness utilizes macroscopic continuous electromagnetic
1357 fields and a “rich alphabet” of diverse molecular and transcriptomic cell states
1358 (Tasic et al., 2018). This allows the coupling matrix A_{ij} to deform across a
1359 continuous parameter space. Our lemma `exists_better_coupling_allocation`,
1360 wrapped as `continuous_beats_rigid_topology` (`Phase7_HardwareComparison.lean`),
1361 formalizes the reallocation step of this argument: whenever a coupling ma-
1362 trix places weight on a less phase-correlated pair while a more correlated pair
1363 exists, there is another matrix with *identical* total coupling resources that
1364 achieves strictly greater phase correlation.

1365 Two objections to that formalization have since been answered, and we
1366 state precisely what is and is not now established. The first was that the real-
1367 location lemma assumes its conclusion: the hypothesis `is_strictly_suboptimal`
1368 — positive weight on a badly-correlated pair while a better one exists — is
1369 most of what one wants to prove about rigid hardware. It is no longer as-
1370 sumed. `Phase7_Rigidity.lean` models an architecture by the set S of site-
1371 pairs it has a *wire* for, and requires its couplings to vanish off S ; weights are
1372 tunable, wiring is not. Then `totalCorrelation_le_of_realizable` bounds
1373 every realizable coupling by $R \cdot M$, where R is the resource budget and M
1374 bounds phase correlation over the wired pairs; the bound is attained; and
1375 `rigid_is_strictly_suboptimal` *derives* the hypothesis whenever the sub-
1376 strate’s best-correlated pair is one the wiring misses. `rigid_gap` makes it
1377 quantitative: an unconstrained reallocation of the same budget gains at least

1378 R times the correlation difference. `Examples.lean` §8 exhibits a three-site
 1379 substrate wired to the one maximally anti-correlated pair, where the gap is
 1380 2. This is resource-matched and no longer question-begging, but note what
 1381 it turns on: fixed wiring support, not continuity. The second objection was
 1382 that nothing in the development distinguished a continuum from a lattice
 1383 — both matrices lived on one finite index type. `Phase7.Rigidity.lean` §3
 1384 gives a distinction that is genuinely typed. In the field functional of Sec-
 1385 tion 7, which integrates a coupling kernel against $\mu \otimes \mu$, an architecture
 1386 with finitely many sites is a `Finset` of points and a field-coupled region
 1387 is a set of positive measure. On a substrate whose measure has no atoms,
 1388 `fieldCorrelation_sited.eq_zero` shows a kernel supported on finitely many
 1389 sites contributes *exactly zero* coupling energy, whatever weights it carries
 1390 and whatever the phase field does, while a patch of strength c on U achieves
 1391 $c\mu(U)^2$. `Examples.lean` §8 instantiates this on the real line: a thousand
 1392 units of coupling concentrated at the origin register zero against a unit patch
 1393 that registers one. The limit of that result is equally plain, and it is not
 1394 resource-matching in the earlier sense — a measure-zero substrate has zero
 1395 resource in the $\mu \otimes \mu$ sense too. What it establishes is that the continuum
 1396 coupling energy is blind to a finitely-sited architecture, which is a statement
 1397 about which functional such an architecture can register in, not a demonstra-
 1398 tion that digital hardware computes worse. Taken together: the resource-
 1399 matched content of the corollary is the wiring-support result, the continu-
 1400 ity content is the measure-theoretic one, and the step from either to “von

1401 Neumann architectures cannot experience unified consciousness” remains an
1402 informal argument.

1403 The remainder of this section extends those two results informally, and
1404 we mark the transition rather than blur it. On the reading we propose, a
1405 GPU’s fixed routing matrix leaves it topologically fragmented: the discrete
1406 layout does not supply the spatial coupling that sheaf-theoretic gluing into
1407 a singular Global Section requires. Without this unified topological space,
1408 rigid hardware cannot undergo the reflexive auto-resonance necessary to map
1409 its own causal boundary and generate a “Self.” The distinction between an
1410 LLM on a GPU and a conscious mind is not the absence of physical thermo-
1411 dynamics in the former, but its structural limitation in achieving topological
1412 integration. This perspective aligns with John Searle’s biological natural-
1413 ism (Searle, 1980; Seth, 2024). Where the Chinese Room argument asserted
1414 that discrete symbol manipulation cannot generate conscious understanding,
1415 our derivation supplies a candidate physical mechanism for the asymmetry
1416 — topological rather than semantic, and, as stated above, formalized only
1417 as far as the wiring-support and measure-theoretic results reach. Asserting
1418 consciousness in fragmented, discrete GPU clusters without a unified phase
1419 transition abandons computational functionalism, relying instead on “silicon
1420 panpsychism”—the assumption that digital computation inherently gener-
1421 ates awareness regardless of topological integration (Goff, 2019).

1422 The same reasoning bears on extreme functionalist thought experiments,
1423 such as a macroscopic system of pulleys operating over millennia to imple-

1424 ment a Large Language Model, or a human manually executing inference
 1425 matrices by hand. Beyond lacking the continuous spatial topology required
 1426 for sheaf-theoretic gluing, these discrete systems violate an implied temporal
 1427 constraint of our theory. Structural resonance requires local oscillators to
 1428 phase-align against the dissipating effects of thermodynamic noise (as mod-
 1429 eled by our critical threshold $K_c = 2D$). A system executing state transitions
 1430 over millennia, or via slow manual calculations, suffers from a terminal tem-
 1431 poral mismatch: its dynamics are far too slow to ever phase-lock into a
 1432 non-equilibrium steady state before thermal noise decoheres the macroscopic
 1433 field. On this account the obstacle to pure symbol manipulation is physi-
 1434 cal rather than semantic: not that the symbols mean nothing, but that the
 1435 substrate carrying them cannot hold a phase-locked field.

1436 **13 Numerical illustrations: methods**

1437 Five figures in this paper are computed rather than drawn, and none of
 1438 them is evidence for a theorem. Where a statement is proved, the figure
 1439 shows what the proved object looks like; where a statement is not proved, the
 1440 figure shows the regime the claim lives in and we say so. Each is produced by
 1441 one script in `simulations/`, named below, and each runs in under a minute
 1442 on a laptop. All five are reproducible exactly: the two quadrature figures are
 1443 deterministic, and the three that draw random numbers fix their seeds.

1444 **Figure 4: the self-consistent order parameter.** The three integrals
1445 $Z(a) = \int_{-\pi}^{\pi} e^{a \cos \theta} d\theta$, $M(a) = \int_{-\pi}^{\pi} \cos \theta e^{a \cos \theta} d\theta$ and $S(a) = \int_{-\pi}^{\pi} \sin^2 \theta e^{a \cos \theta} d\theta$
1446 are evaluated by the trapezoidal rule on 4001 points over the same interval
1447 the Lean definitions integrate over, with the weight computed as $e^{a \cos \theta - |a|}$
1448 so that the common factor cancels in the ratios and nothing overflows at
1449 large concentration. The coherent branch is obtained by bisecting $E(a) =$
1450 $S(a)/Z(a)$ against D/K over $a \in [10^{-9}, 200]$ for 200 iterations, then reading
1451 $r = aD/K$; bisection is valid here because E is proved strictly decreasing.
1452 Two checks are printed by the script rather than only plotted: the computed
1453 branch is strictly increasing at every one of its 500 sampled couplings, and
1454 the residual $|r - R(Kr/D)|$ at $K = 4D$ is 2×10^{-16} . Script: `bifurcation.py`.

1455 **Figure 5: the finite- N transition.** $N = 100$ oscillators, natural fre-
1456 quencies drawn from a Lorentzian of scale $D = 1$ (clipped at $\pm 10D$ so
1457 the solver does not stiffen), uniform random initial phases, integrated with
1458 `scipy.integrate.odeint` over $t \in [0, 100]$ on 1000 steps for each of 20 cou-
1459 plings in $[0, 4D]$. The order parameter is $|N^{-1} \sum_j e^{i\theta_j}|$ averaged over the last
1460 100 steps. This is the deterministic quenched-disorder model, not the noisy
1461 identical-frequency model of the Lean development; the caption records why
1462 both have threshold $2D$. Script: `kuramoto.py`.

1463 **Figure 2: mesh refinement.** The kernel is $K(x, y) = e^{-d(x, y)^2/2}$ with d
1464 the periodic distance on the ring of circumference 2π , the phase field is $\theta(x) =$
1465 $\sin x$, and the potential is $-\frac{1}{2} \iint K \cos(\theta_y - \theta_x)$. The continuous value is a

1466 2000-point Riemann sum standing in for the exact integral; the discrete values
 1467 use $N = 10, 20, \dots, 640$ nodes with edge weights $K(x_i, x_j)(2\pi/N)^2$. Plotted
 1468 is the absolute error against an $\mathcal{O}(N^{-2})$ reference. Because the “exact”
 1469 value is itself a 2000-point sum, the last data point ($N = 640$) is within a
 1470 factor of ten of the reference computation and should be read as such. That
 1471 $\mathcal{O}(N^{-2})$ reference line is no longer only empirical: on the formalized one-
 1472 dimensional grid the discrete energy is exactly the composite trapezoidal rule
 1473 (`grid_discreteEnergy_eq_trapezoidal`), so for a C^2 integrand with second
 1474 derivative bounded by ζ the error is at most $\zeta/(12N^2)$ (`grid_energy_error_`
 1475 `le`), a bound attained at $N = 1$ for $y \mapsto y^2$. Script: `mesh_refinement.py`.

1476 **Figure 3: structural resonance.** $N = 50$ units, 2000 steps of forward
 1477 Euler at $dt = 0.05$. At each step the drive is a fresh draw from $\mathcal{N}(0, \Sigma_{\text{ext}} +$
 1478 $0.1I)$ with Σ_{ext} a fixed normalized rank-5 covariance; the phases advance by
 1479 $\dot{\theta}_i = \sum_j K_{ij} \sin(\theta_j - \theta_i) + \xi_i$; entropy production is recorded as $\sum_i \dot{\theta}_i^2$; and
 1480 the coupling descends its gradient with rate $\eta = 0.01$, symmetrized, clipped
 1481 to $[0, 1]$ and held hollow. The left panel is smoothed with a 50-step boxcar.
 1482 Note that the plotted σ is the squared drift without the $1/D$ prefactor of the
 1483 functional in Section 10, which is a constant rescaling and changes nothing
 1484 about the shape. Script: `structural_resonance.py`.

1485 **Figure 6: rigid against deformable coupling.** $N = 30$, 1000 steps of
 1486 forward Euler at $dt = 0.05$, shared initial phases and frequencies. The rigid
 1487 matrix is a symmetric random matrix thresholded at 0.7 and normalized

1488 to $\sum_{ij} A_{ij} = 2N$; the adaptive copy starts from the same matrix, ascends
1489 $-\partial V/\partial A_{ij}$ at rate $\eta = 0.1$, is clipped non-negative and hollow, and is renor-
1490 malized to the same total after every step, which is what makes the com-
1491 parison resource-matched. Both potentials are $V = -\frac{1}{2} \sum_{ij} A_{ij} \cos(\theta_j - \theta_i)$.
1492 Script: `hardware_comparison.py`.

1493 14 Conclusion

1494 Human-like cognitive architectures do not require biological mysticism or
1495 exotic sub-quantum physics (Hameroff & Penrose, 2014). While quantum
1496 theories of consciousness often invoke delicate subatomic phenomena to ex-
1497 plain subjective unity, every step of the chain assembled here is classical:
1498 broken symmetries, finite phase space, Landauer erasure, and gradient de-
1499 scent on entropy production. What the formalization establishes is that this
1500 chain is mathematically coherent and that its remaining physical commit-
1501 ments are localized and named — not that classical fields are *sufficient* for
1502 consciousness. Two links carry that weight, and their status is no longer the
1503 same. The unified state is proved to glue uniquely once a cover sits at the
1504 minimum of the coupling potential; the motion is proved to stop along every
1505 trajectory (`velocity_sq_tendsto_zero`); and *where* it stops is now proved,
1506 conditionally — for initial data within a quarter turn and below an explicit
1507 energy threshold, the trajectory converges to a global minimiser of the cou-
1508 pling potential (`kuramoto_tendsto_global_minimum`), and a cover built on

1509 that limit discharges the equilibrium hypothesis Derivation 5 used to assume
1510 (`ThermodynamicCover.ofConvergentTrajectory`). What is *not* proved is
1511 the unconditional statement, and it is false: splay and twisted configurations
1512 are equilibria of the same flow, so the basin condition is the scope of the claim
1513 rather than slack in it. That a physical cortex starts inside that basin is an
1514 assumption. The second link is untouched by any of this: the identification
1515 of a global section of the probability sheaf with the unity of experience is a
1516 stipulation of the framework, not a consequence of it. Subject to those two,
1517 cognitive architectures can be modeled as macroscopic attractor states of a
1518 universe governed by broken symmetries, finite phase space, and the physical
1519 imperative to minimize thermodynamic dissipation.

1520 Philosophically, this framework bridges the historical gap between Type
1521 Identity Theory (Smart, 1959) and Computational Functionalism (Putnam,
1522 1967). The subjective mental state is identical to the physical thermody-
1523 namic field (Identity Theory), yet its qualitative nature is defined by the
1524 topological and functional dynamics of structural resonance rather than the
1525 specific carbon-based biochemistry of the brain (Functionalism). This con-
1526 strains the *multiple realizability* of mind: consciousness is not exclusively
1527 bound to biological carbon, but neither is it substrate-neutral, because the
1528 substrate is what fixes which couplings are physically available. How far that
1529 constraint reaches is the question the corollary answers only in part. What
1530 is proved is that fixed wiring is strictly suboptimal at matched resource, and
1531 that a finitely-sited architecture registers exactly zero coupling energy in the

continuum functional; the step from either to “von Neumann architectures cannot be conscious” is an argument we make in prose and do not formalize.

By anchoring these derivations in Russellian Monism, this framework acknowledges the constitutive limits of third-person scientific explanation. As argued by Love (Love, 2026), building upon Nagel’s assertion that subjective phenomenal experience is inherently not amenable to objective measurement (Nagel, 1986), the “hard problem” of consciousness may represent a category error rather than a solvable scientific puzzle. Science can map the physical dynamics of topological resonance and sheaf-theoretic gluing, but it is structurally incapable of adjudicating the first-person qualitative experience. Our formalization respects this boundary: the Lean 4 theorems verify the objective physical attractors, while the intrinsic “what it is like” to be that resonant field remains the domain of the first-person perspective. By locating the phase transition at a threshold with a closed-form value — $K_c = 2D$ for the noisy mean-field Kuramoto model the framework adopts, conditional on the von Mises stationary density and not yet linked to a trajectory — it aligns with what Anil Seth terms the “real problem” of consciousness (Seth, 2021). Rather than attempting to bridge an unbridgeable metaphysical gap, it offers a topological architecture that can help explain, predict, and physically perturb the structural properties of experience.

Future empirical validation of this thermodynamic framework may leverage 3D cortical organoids, offering a controlled biological environment to systematically manipulate metabolic substrates and observe the topological

1555 emergence of these coherent phase transitions.

1556 **Code Availability**

1557 The complete Lean 4 formalization, encompassing the derivation of all theo-
1558 rems presented in this manuscript, as well as the accompanying Python sim-
1559 ulations, are publicly available in the GitHub repository: [https://github.](https://github.com/bobaseb/physical_primitives_of_mind)
1560 [com/bobaseb/physical_primitives_of_mind](https://github.com/bobaseb/physical_primitives_of_mind). Additional documentation
1561 and web-accessible proofs are hosted at the project website: [https://bobaseb.](https://bobaseb.github.io/physical_primitives_of_mind/)
1562 [github.io/physical_primitives_of_mind/](https://bobaseb.github.io/physical_primitives_of_mind/). Each of the five computed
1563 figures is produced by one script in `simulations/` and regenerates exactly,
1564 Section 13 giving the parameters; Figure 1 is drawn rather than computed.

1565 **Contributions**

1566 The conceptualization, manuscript drafting, and Lean 4 formalizations pre-
1567 sented in this work were heavily AI-assisted, and we report which tools
1568 did what rather than acknowledging them in general terms. The Google
1569 Antigravity CLI, running Gemini 3.1 and Claude 4.6 Sonnet, served as a
1570 collaborative reasoning partner throughout the research and formalization.
1571 DeepSeek V4 Flash and Claude Opus 5 (routed via OpenRouter) operat-
1572 ing through Nous Research’s Hermes Agent framework provided substantial
1573 assistance with code generation, proof debugging, and structural analysis.

1574 Claude Opus 5 running in Claude Code carried out the later formalization
1575 passes — the domain-wall theorem, the thermodynamics-of-prediction re-
1576 placement, the equilibrium bridge — together with the figures and the re-
1577 structuring of this manuscript. Responsibility for every claim, and for the
1578 decision about what each machine-checked result does and does not establish,
1579 rests with the author.

1580 References

- 1581 Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable
1582 electric fields emerge from and allow representational drift. *NeuroImage*,
1583 253, 119058.
- 1584 Bauer, L. G., et al. (2022). Quantification of Kuramoto Coupling Between
1585 Intrinsic Brain Networks Applied to fMRI Data in Major Depressive Dis-
1586 order. *Frontiers in Computational Neuroscience*, 16.
- 1587 van Bree, S., et al. (2025). Processes and measurements: a framework for
1588 understanding neural oscillations in field potentials. *Trends in Cognitive*
1589 *Sciences*.
- 1590 Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated informa-
1591 tion theory: from consciousness to its physical substrate. *Nature Reviews*
1592 *Neuroscience*, 17(7), 450-461.

- 1593 Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the
1594 Mind. *Oxford University Press*.
- 1595 Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a compu-
1596 tational theory of conscious processing. *Current Opinion in Neurobiology*,
1597 25, 76-84.
- 1598 Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature*
1599 *Reviews Neuroscience*, 11(2), 127-138.
- 1600 Kuramoto, Y. (1984). Chemical Oscillations, Waves, and Turbulence.
1601 *Springer Series in Synergetics*, Vol. 19.
- 1602 Scherf, M. (2026). A Formal Proof of Non-Duality: An Exposition of the
1603 'OnlyOne' System. *Unpublished manuscript, PhilPeople*.
- 1604 Inoué, T. (2026). On Brain as a Mathematical Manifold: Neural Manifolds,
1605 Sheaf Semantics, and Leibnizian Harmony. *arXiv:2601.15320 [q-bio.NC]*.
- 1606 Spisák, T., & Friston, K. (2026). Self-orthogonalizing attractor neural net-
1607 works emerging from the free energy principle. *Neurocomputing*.
- 1608 Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
1609 memory network formation. *Cerebral Cortex*, 33(17), 9877-9890.
- 1610 Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
1611 Electric fields sculpt neural activity and "tune" the brain's infrastructure.
1612 *Progress in Neurobiology*, 226, 102465.

1613 Ruffini, G., Salvador, R., Tadayon, E., Sanchez-Todo, R., Pascual-Leone, A.,
1614 & Santarnecchi, E. (2020). Realistic modeling of mesoscopic ephaptic cou-
1615 pling in the human brain. *PLOS Computational Biology*, 16(6), e1007923.

1616 Cunha, G. M., Corso, G., de Sousa, M. P. B., & Lima, G. Z. d. S. (2024).
1617 Can ephapticity contribute to brain complexity?. *PLOS ONE*, 19(12),
1618 e0310640.

1619 Goldwyn, J. H., & Rinzel, J. (2016). Neuronal coupling by endogenous elec-
1620 tric fields: Cable theory and applications to coincidence detector neurons
1621 in the auditory brainstem. *Journal of Neurophysiology*, 115(4), 2033-2051.

1622 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
1623 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
1624 9(8), 45-60.

1625 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
1626 testable hypothesis about the characteristics of conscious as opposed to
1627 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.

1628 Goff, P. (2019). *Galileo's Error: Foundations for a New Science of Con-*
1629 *sciousness*. Pantheon Books.

1630 Seth, A. (2021). *Being You: A New Science of Consciousness*. Penguin.

1631 Seth, A. (2024). Conscious artificial intelligence and biological naturalism.
1632 *PsyArXiv*.

- 1633 Levin, M. (2019). The computational boundary of a “self”: developmental
1634 bioelectricity drives multicellularity and scale-free cognition. *Frontiers in*
1635 *Psychology*, 10, 2688.
- 1636 Friston, K., & Kiebel, S. (2009). Predictive coding under the free-energy
1637 principle. *Philosophical Transactions of the Royal Society B: Biological*
1638 *Sciences*, 364(1521), 1211-1221.
- 1639 Clark, A. (2016). *Surfing Uncertainty: Prediction, Action, and the Embodied*
1640 *Mind*. Oxford University Press.
- 1641 Hohwy, J. (2013). *The Predictive Mind*. Oxford University Press.
- 1642 Prigogine, I., & Stengers, I. (1984). *Order Out of Chaos: Man’s New Dialogue*
1643 *with Nature*. Bantam Books.
- 1644 McFadden, J. (2020). Integrating information in the brain’s EM field: the
1645 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
1646 niaa016.
- 1647 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
1648 guide neocortical network activity. *Neuron*, 67(1), 129-143.
- 1649 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
1650 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
1651 95-103.

- 1652 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
1653 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.
- 1654 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
1655 on brain circuits in rats and humans. *Nature Communications*, 9, 483.
- 1656 Tasic, B., et al. (2018). Shared and distinct transcriptomic cell types across
1657 neocortical areas. *Nature*, 563(7729), 72-78.
- 1658 Verkhratsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
1659 *logical Reviews*, 98(1), 239-389.
- 1660 Picard, M., & Shirihai, O. S. (2022). Mitochondrial signal transduction. *Cell*
1661 *Metabolism*, 34(11), 1620-1638.
- 1662 Harnad, S. (1990). The symbol grounding problem. *Physica D: Nonlinear*
1663 *Phenomena*, 42(1-3), 335-346.
- 1664 Tononi, G., & Koch, C. (2015). Consciousness: here, there and everywhere?
1665 *Philosophical Transactions of the Royal Society B: Biological Sciences*,
1666 370(1668), 20140167.
- 1667 Aaronson, S. (2014). Why I Am Not An Integrated Information Theorist
1668 (or, The Unconscious Expander). *Shtetl-Optimized*. Retrieved from [https:](https://scottaaronson.blog/?p=1799)
1669 [//scottaaronson.blog/?p=1799](https://scottaaronson.blog/?p=1799).
- 1670 Block, N. (1995). On a confusion about a function of consciousness. *Behav-*
1671 *ioral and Brain Sciences*, 18(2), 227-247.

- 1672 Herculano-Houzel, S. (2009). The human brain in numbers: a linearly scaled-
1673 up primate brain. *Frontiers in Human Neuroscience*, 3, 31.
- 1674 Lange, W. (1975). Cell number and cell density in the cerebellar cortex of
1675 man and some other mammals. *Cell and Tissue Research*, 157(1), 115-124.
- 1676 Yu, F., Jiang, Q. J., Sun, X. Y., & Zhang, R. W. (2015). A new case of
1677 complete primary cerebellar agenesis: clinical and imaging findings in a
1678 living patient. *Brain*, 138(6), e353.
- 1679 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
1680 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
1681 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.
- 1682 Hameroff, S., & Penrose, R. (2014). Consciousness in the universe: A review
1683 of the 'Orch OR' theory. *Physics of Life Reviews*, 11(1), 39-78.
- 1684 Hunt, T., & Schooler, J. W. (2019). The easy part of the hard problem: a
1685 resonance theory of consciousness. *Frontiers in Human Neuroscience*, 13,
1686 378.
- 1687 England, J. L. (2018). Dissipative adaptation in driven self-assembly. *Nature*
1688 *Nanotechnology*, 13(4), 287-295.
- 1689 Ganesh, N., & Parrondo, J. M. R. (2021). Thermodynamic computing with
1690 non-equilibrium systems. *Nature Physics*.

- 1691 Hinton, G. (2022). The Forward-Forward Algorithm: Some Preliminary In-
1692 vestigations. *arXiv preprint arXiv:2212.13345*.
- 1693 Searle, J. R. (1980). Minds, brains, and programs. *Behavioral and Brain*
1694 *Sciences*, 3(3), 417-424.
- 1695 Smart, J. J. C. (1959). Sensations and brain processes. *The Philosophical*
1696 *Review*, 68(2), 141-156.
- 1697 Putnam, H. (1967). Psychological predicates. In *Art, Mind, and Religion*.
1698 University of Pittsburgh Press.
- 1699 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
1700 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
1701 brain. *Science*, 342(6156), 373-377.
- 1702 Nagel, T. (1986). *The View from Nowhere*. Oxford University Press.
- 1703 Love, B. C. (2026). Consciousness, AI, and the Limits of Scientific Explana-
1704 tion. *arXiv preprint arXiv:2606.00226*.
- 1705 Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return
1706 of Maxwell's demon. *Studies in History and Philosophy of Science Part B:*
1707 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.
- 1708 Landauer, R. (1961). Irreversibility and heat generation in the computing
1709 process. *IBM Journal of Research and Development*, 5(3), 183-191.

- 1710 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
1711 *of Physics A: Mathematical and General*, 9(8), 1387-1398.
- 1712 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
1713 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.
- 1714 Barbălat, I. (1959). Systèmes d'équations différentielles d'oscillations non
1715 linéaires. *Revue Roumaine de Mathématiques Pures et Appliquées*, 4, 267-
1716 270.
- 1717 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
1718 of prediction. *Physical Review Letters*, 109(12), 120604.
- 1719 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
1720 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.
- 1721 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
1722 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
1723 181.
- 1724 Coddington, E. A., & Levinson, N. (1955). Theory of Ordinary Differential
1725 Equations. *McGraw-Hill*, New York.
- 1726 Cover, T. M., & Thomas, J. A. (2006). Elements of Information Theory (2nd
1727 ed.). *Wiley-Interscience*, Hoboken, NJ.
- 1728 Shenker, O. R. (2000). Logic and entropy. *Philosophy of Science*, 67, S146-
1729 S152.